FlowTrac-II

Versatile Pressure and Volume Control Flow Pump System for Running Fully Automated Triaxial (all types), Hydraulic Conductivity (Permeability), CRC and Rowe Consolidation Tests

Using Windows® 10/11

Geocomp
A SERCEL COMPANY

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978-635-0012

User's Manual

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SECTION 1: OVERVIEW

FlowTrac is an intelligent, versatile unit composed of a flow pump, a pressure sensor and a control board. The flow pump, which regulates pressure and volume in a test cell, contains a high speed, precision micro stepper motor that moves a piston in a water-filled cylinder. An embedded control with a dedicated CPU uses readings from the pressure transducer mounted at the end of the cylinder to determine what signals should be sent to the stepper motor.

The number of motor steps is used to calculate flow pump volume change. Two two-way solenoid valves are used to control the direction of flow. The output valve controls whether or not pressure can be applied to a test cell. The supply valve controls whether or not the fill/drain operation can occur. (For normal operation, when one valve is open, the other is closed.)

FlowTrac is capable of maintaining a desired pressure within 0.35 kPa (0.05 psi) while monitoring volume changes within 0.001 cc. Pressure increments may increase and decrease in any pattern by any amount (as long as the system's limits are not exceeded) as specified by the user, even while a test is in progress.

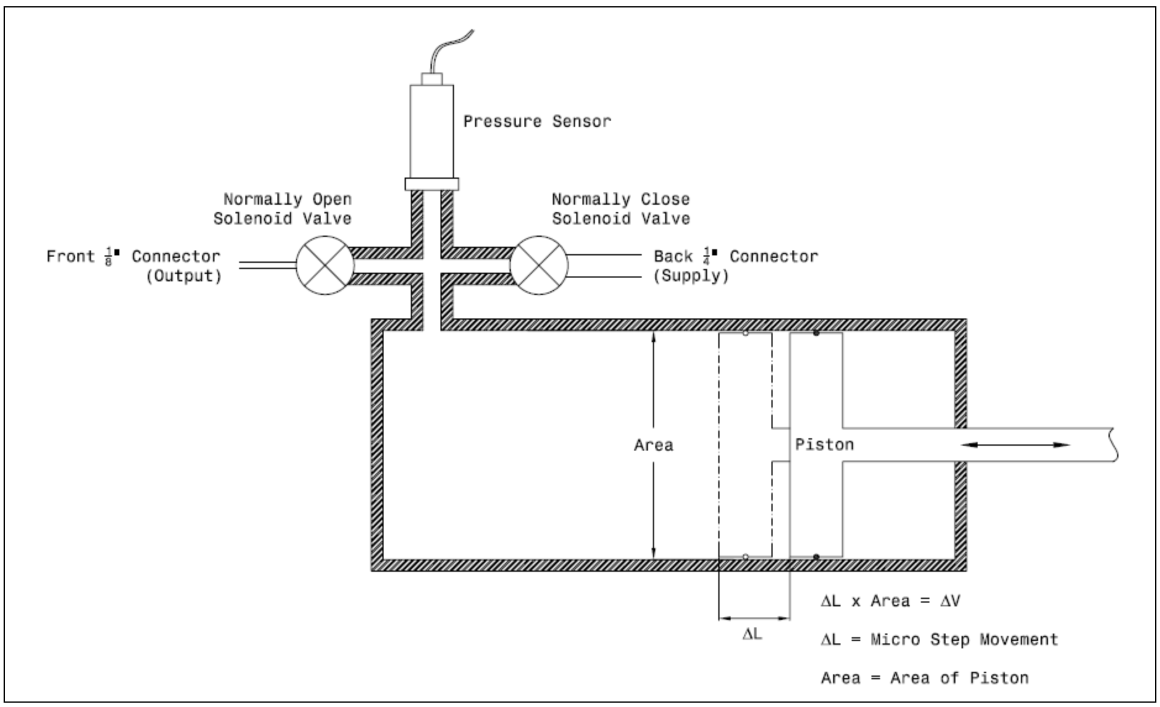


Figure 1. Schematic of a FlowTrac Unit

You can manually control FlowTrac by using a keypad and an LCD screen that displays easy-to-use menus. These are on the front panel of the unit.

The FlowTrac system is composed of one or more FlowTrac units and a computer with monitor to set up, control and monitor a test. Specially designed software allows you to define the conditions for running a particular test. The information is entered by way of intuitive entry windows. The software also will produce a report of the test results in tabular and graphical form.

The system requires no special skills to operate, other than those used in conventional testing. A person familiar with soil testing can learn to use the system confidently within a few days. Familiarity with using Microsoft Windows can significantly reduce the learning time.

1.1. FlowTrac System Hardware

The FlowTrac hardware consists of the following parts:

- **FlowTrac flow pump:** Uniquely designed unit that contains a flow pump, an embedded controller and a pressure sensor. These components control and measure pressure and volume change in a test cell. An LCD screen and a keypad on the front panel allow you to manually control the operation of the flow pump and monitor the system status.
- **Test accessories:** Triaxial cell, permeability cell, CRC cell, or ROWE cell.
- **Computer**: PC computer running Windows XP, Vista or 7 with a Network/Communication card installed.
- **Keyboard and mouse:** Standard keyboard and mouse for computer control.
- **Display:** Computer monitor.

Figure 2 and Figure 3 show FlowTrac construction details. Figure 4 shows the FlowTrac system as it would be used to perform a permeability test.

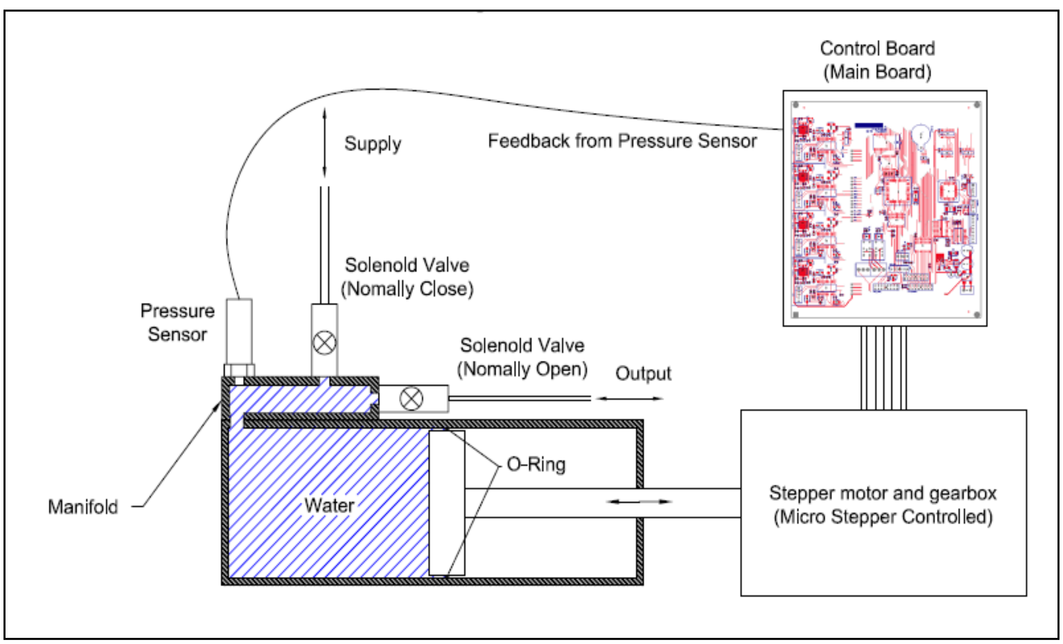


Figure 2. FlowTrac Diagram

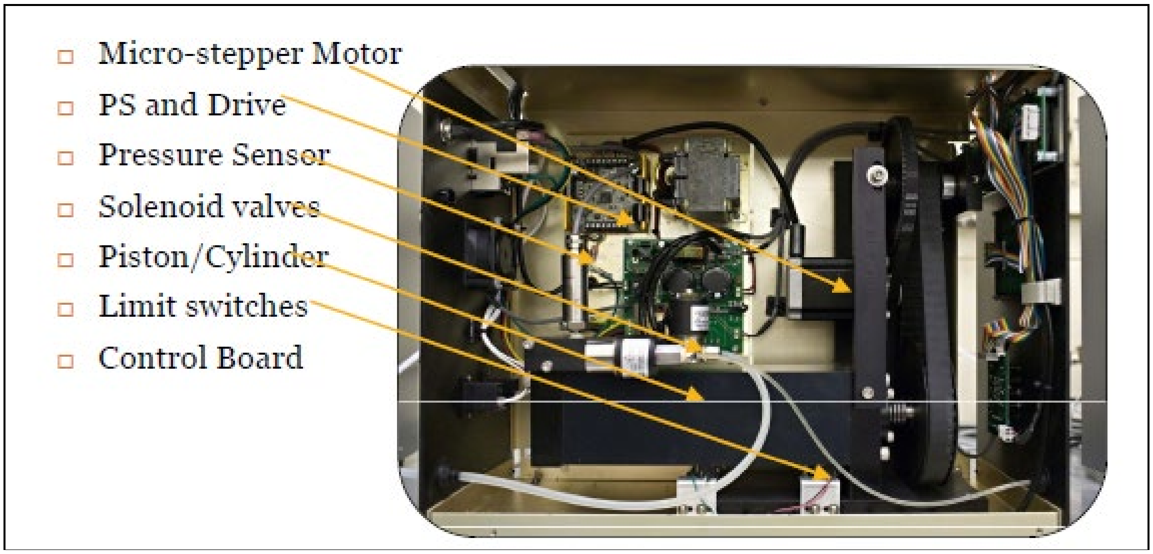


Figure 3. Inside the FlowTrac Frame

Figure 4. FlowTrac II System for Permeability (Hydraulic Conductivity) Testing

1.2. Principles of Operation

While a stand-alone FlowTrac unit can be used to perform certain basic test functions, the FlowTrac system (that includes a computer and usually a LoadTrac-II load frame) operates as an intelligent control system that can automate many different types of tests. The computer, by way of the installed interface card and specially designed software, controls one, two or three FlowTrac units as well as the LoadTrac-II unit. The FlowTrac system can be used to provide constant pressure, constant rate of pressure change, constant volume, constant rate of volume change, or any combination thereof.

The specially designed software packages allow you to set all the parameters that are required for running a test.

The computer uses feedback from the pressure transducer in a FlowTrac unit to provide real-time control of pressure or volume change. The computer sends commands to the imbedded controller which in turn generates signals to run the stepper motor. The stepper motor is connected to a ball screw that moves a piston in a water-filled cylinder. By moving the piston into the cylinder, the pressure in the water increases and water flows from the cylinder to the test chamber. By moving the piston out of the cylinder, the pressure in the water decreases and water flows from the test chamber into the cylinder. The piston is moved until the motor step count and/or the reading from the pressure transducer equal the values required to meet the test specifications. FlowTrac is designed so that one step of the stepper motor on a 250 cc unit corresponds to a volume change of about 5.4 x 10^{-4} cc and on a 750 cc unit to about 1.7 x 10^{-3} cc. The following equations are used to convert steps to volume change.

250 cc unit: $\Delta V(cc) = (5.4x10^{-4})(\Delta steps)$

750 cc unit: $\Delta V(cc) = (1.7x10^{-3})(\Delta steps)$

Each FlowTrac unit is configured at the factory with a highly accurate, electrically isolated pressure sensor. This pressure sensor features a silicon piezoresistive-sensing element. This technology provides exceptional accuracy, stability, repeatability, and immunity from temperature changes. The pressure sensor connects to the embedded controller. The controller supplies excitation power to the pressure sensor and conditions the sensor's analog output signal before converting it to digital information. The digitized signal is then sent to the computer that is running the test control software. It is also converted to engineering units, and is viewable on the flow pump front panel LCD (System Monitor option).

The FlowTrac embedded controller uses firmware-controlled signal conditioning to filter and amplify the low-level pressure transducer analog output and change it to a voltage range which spans 5 volts (-2.5 to +2.5 V). This high-level signal is sent to a 16-bit A/D converter. The result is that the transducer reading is converted to an integer between 0 and 65536 (0 and 2^{16}). An integer value generated by the A/D converter is referred to as the number of counts (cnts). The pressure transducer has an analog output that spans a voltage range of no more than 320 mV (-160 mV to +160 mV). The net effect of the conversion to digital form is that each count corresponds to about 0.005 mV.

An adjustable offset in the embedded control system allows a specific count reading to be set for the "zero" condition of the pressure transducer. The system is configured so that a reading of about 30000 cnts for the pressure transducer (set as bipolar) corresponds to atmospheric pressure. The full-scale output for the transducer is near 65000 cnts.

The count readings must be converted to engineering units. Each unit is shipped with a calibration factor that provides this conversion. Linear equations are used to relate engineering units to count readings. For example, a 2000 kPa (300 psi) pressure transducer is typically set up so that this maximum pressure on it gives an output reading of approximately 65000 cnts. Since atmospheric pressure corresponds to about 30000 cnts, the 2000 kPa (300 psi) pressure spans a range of about 35000 cnts. Consequently, the calibration factor would be approximately 2000 kPa/35000 cnts, or about 0.06 kPa/cnt (0.009 psi/cnt). Consequently, one cnt corresponds to about 0.06 kPa (0.009 psi). The software for running a test with FlowTrac includes an easy to use calibration feature that simplifies a transducer calibration so that you can get an exact calibration factor with very little effort.

All components must be functioning correctly for the FlowTrac system to operate. To avoid damage to any part of the system, readings from all sensors are checked hundreds of times per second. The detection of any analog or digital reading outside the acceptable range will prevent the stepper motor from running. Consequently, if a transducer becomes disconnected or defective, FlowTrac will not run.

1.3. FlowTrac Software

The software that automates a test being run with the FlowTrac system also contains a sensor calibration feature and an editing/reporting feature. The control feature of the program runs the test, collects data for the test, and stores the data in a file on the hard drive.

The sensor calibration feature calculates a calibration factor and offset for a sensor on the basis of a standard (calibrated pressure gauge for pressure, gauge blocks for a displacement transducer if one is needed on channel 2) after you have entered sensor readings into a table.

The editing/reporting feature reads the data from the test file, performs the necessary calculations, and prepares the final test result tables and graphs for a report.

The control feature of the software can be run only on a computer that has the Geocomp Network/Control card installed on it. The editing/reporting feature may be run on any computer with Windows XP(SP3), 7 or 10.

The software is supplied on a CD shipped with your system. Instructions for installing the software (and the Network/Controller card) on a computer are included with the FlowTrac system shipment (and also included on the CD as backup).

SECTION 2: USE OF FLOWTRAC

These instructions assume that your FlowTrac system has been installed following the instructions in Appendix A Installation and is ready for operation. Should you have any difficulty with the operation instructions, consult Appendix C Troubleshooting or call Geocomp, Inc. for assistance.

Turn on your computer and monitor (if they are not already on). Open a control program that you will use for running a test.

Turn on the FlowTrac unit. The imbedded controller will go through a boot sequence which will take about fifteen seconds.

WARNING: **Do not press the Esc key on the keypad during the boot sequence. Your flow pump firmware has been configured at the factory for the conditions of your use. Pressing the Esc key during the boot sequence will remove this configuration.**

While the control system is booting the Motor Power and CPU Power lights will glow steadily while the Flow Out and Flow In lights will alternate rapidly on and off. Toward the end of the boot sequence, the Net Tx light will come on. All three LEDs in the left column must be a steady green for FlowTrac to function properly.

NOTE: **If the Net Tx LED is blinking, the network cable used to connect your computer to the load frame is probably not connected properly or a control program has not been opened. You may have to load a test file into your control program before the Net Tx LED will glow steadily. See your software manual for loading details.**

If the Net Tx LED is still blinking even though a control program is running (or one or more of the other LEDs is not glowing steadily), refer to Appendix C Troubleshooting or call Geocomp, Inc. for assistance.

At the completion of the boot sequence, the main menu (Figure 5) will appear on the LCD screen.

Figure 5. Main Menu

The ID number should be 66 if your application requires only one FlowTrac unit. (If your application requires two units, one should have the ID of 66 and the other 67 depending how the units are to be used. If your application requires three units, the IDs should be 66, 67 and 68.)

If you are using a power strip with a switch, you can use that switch to simultaneously turn on and off the entire FlowTrac system. But the Net Tx LED may not glow steadily until a control program is loaded.

2.1. Manual Operation of the FlowTrac Unit

The keypad located on the front panel is used to select options and enter parameters that are used in the manual control of the FlowTrac unit. The keypad layout is shown below.

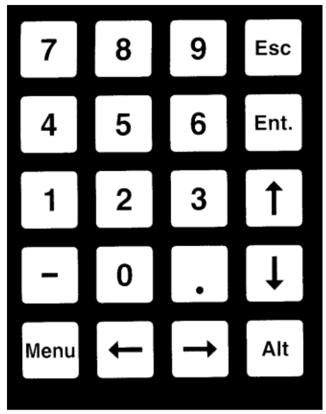


Figure 6. Keypad

Options from the main menu and other menus are chosen by pressing the appropriate numeric key. From any other screen display, the main menu can be shown by pressing the Menu key. From any other screen display, the previous display can be shown by pressing the Esc key. Additional key use will be explained in the sections below.

The complete menu tree is shown in Figure 7.

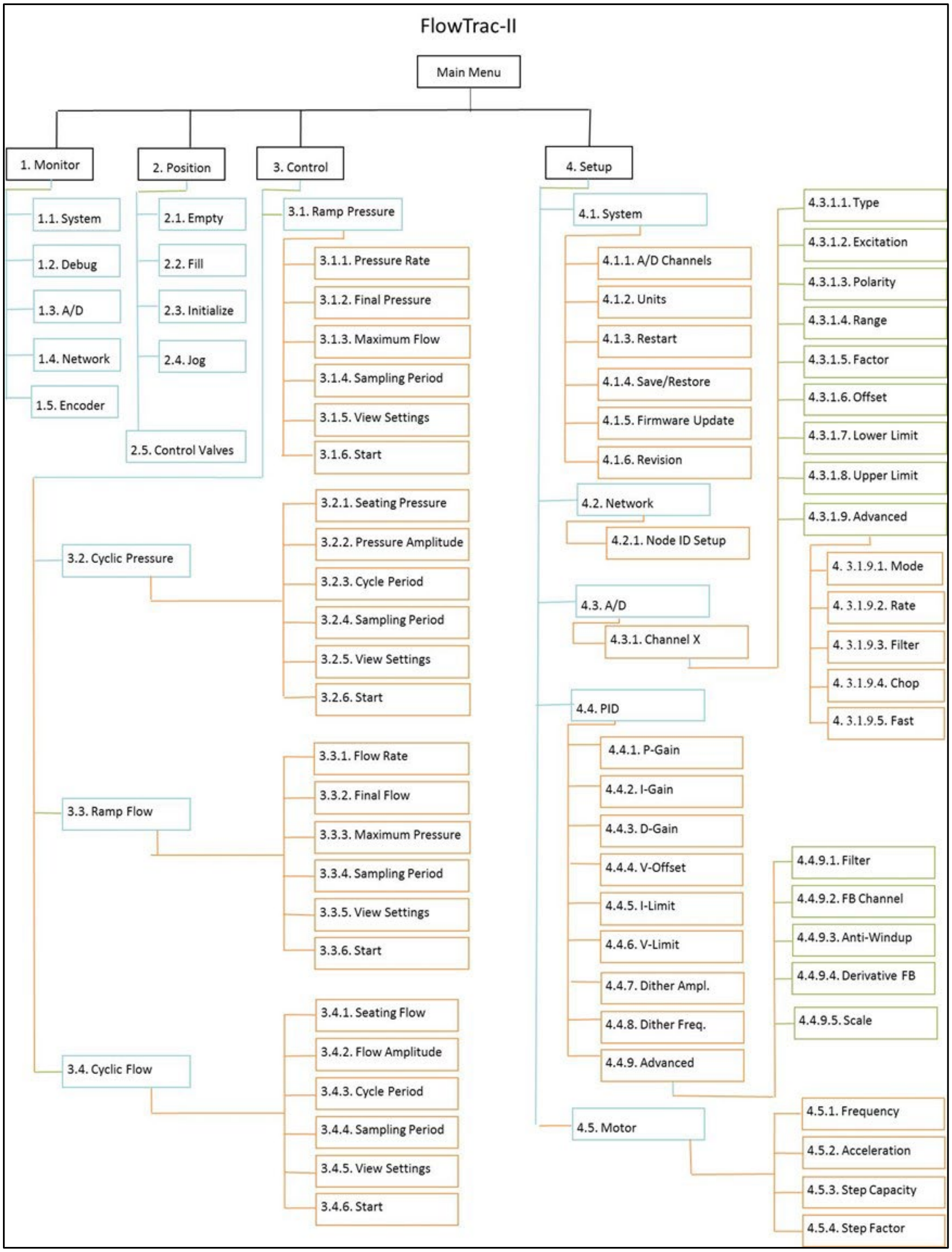


Figure 7. Menu Tree for FlowTrac

2.1.1. Monitor

With the main menu displayed on the LCD screen, select the Monitor option by pressing 1 on the keypad. The Monitor menu (Figure 8) will appear on the LCD screen.

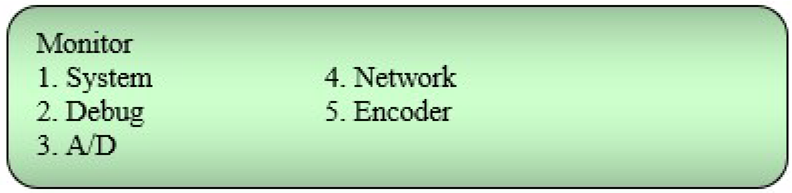


Figure 8. Monitor Menu

Select the System option to display the current conditions for the FlowTrac unit. These include the internal pressure transducer reading (specified on the display as Press for pressure) in engineering units (either metric or English), the volume of water contained in the flow pump cylinder (in cc), how full the cylinder is (in percent), the status of the empty and full limit switches, and the status of the output and supply valves. If a second pressure sensor is being used, its reading (specified on the display as Pore) is also displayed in engineering units.

Select Debug to display detailed information about the functioning of the FlowTrac unit. This option and the A/D option will be needed only if the FlowTrac system is not functioning correctly, in which case you will need to consult a technician at Geocomp, Inc.

Select Network to display the network information. Four columns of four numbers each will be shown. The top number of the first column is the ID for the flow pump. The second and third numbers are how many communication packets have been transferred between your computer and the load frame since the last load frame boot. The fourth number is the transfer rate per second. The rest of the numbers indicate the status of various functions of the system and are used for diagnostic purposes.

Select Encoder to display encoder information. There will be two columns of numbers with two numbers in each column. In addition, a running time will be displayed at the lower right of the screen. This display shows the time that has elapsed since the system was booted.

2.1.2. Position

With the main menu displayed on the LCD screen, select the Position option by pressing 2 on the keypad. The Position menu (Figure 9) will appear on the LCD screen.

Figure 9. Position Menu

The first four options on the Position menu allow you to control the motion of the flow pump piston and so set its position. The fifth option allows you to open or close the valves.

***WARNING: Unless it is absolutely necessary, do not move the piston in the flow pump cylinder (by using the Fill, Empty, Initialize or Jog options) unless there is water in the cylinder. Dry operation can damage the seal around the piston. The fifth option is used to set the status of the Output Valve and the Supply Valve.**

2.1.2.1. Empty

To empty the flow pump cylinder, select the Empty option on the Position menu, then press Ent. The yellow "Output Valve" LED will glow, indicating that the output valve is open, and the green "Flow Out" LED will go on. The pump motor will start running, and the flow pump cylinder will empty at the rate determined by the frequency setting and motor step factor, typically about 8.7 for the 250 cc unit and about 28 cc/sec for the 750 cc unit cc/sec. You can stop the motion by pressing any key on the keypad. While the pump motor is running, the value of the step count (shown on the LCD screen) will increase. Also displayed on the LCD screen will be the reading of the sensors in engineering units. The pump will continue to empty until the pump piston reaches the Empty limit switch. When the Empty limit switch is triggered, the motor will automatically stop running, the Step Count will be set to the Step Capacity value (see Motor Setup in Section 2.1.4.5 Motor), the bottom line of the LCD display will read "Pump is empty," and the red "Limit Empty" LED on the front panel will blink.

2.1.2.2. Fill

***CAUTION: When using the Fill option, be sure the tubing connected to the supply port is in a container of water so the pump cylinder does not fill with air.**

To fill the flow pump cylinder, select the Fill option on the Position menu, then press Ent. The yellow "Supply Valve" LED will glow, indicating that the supply valve is open, and the green "Flow In" LED will go on. The pump motor will start running, and the flow pump cylinder will fill at the rate determined by the frequency setting and motor step factor, typically about 8.7 for the 250 cc unit and about 28 cc/sec for the 750 cc unit cc/sec. You can stop the motion by pressing any key on the keypad. While the pump motor is running, the value of the step count (shown on the LCD screen) will decrease. Also displayed on the LCD screen will be the reading of the sensors in engineering units. The pump will continue to fill until the pump piston reaches the Full limit switch. When the Full limit switch is triggered, the motor will automatically stop running, the Step Count will be set to zero, the bottom line of the LCD display will read "Pump is full," and the red "Limit Full" LED on the front panel will blink.

2.1.2.3. Initialize

To initialize the flow pump, select the Initialize option on the Position menu, and then press Ent. The flow pump cylinder will begin to empty, and the behavior of the FlowTrac unit will be the same as when you choose the Empty option. However, as soon as the Empty limit switch is reached and the Step Count is set to zero, the motor will reverse its direction. The green "Flow In" LED will go on, and the flow pump cylinder will fill until it is half full. This initialization option should be used only when the FlowTrac unit is not being controlled by a computer. When an automated test is being run, initialization should be done by way of the control program.

2.1.2.4. Jog

The Jog option allows you to move the flow pump piston in either direction and at any desired motor step rate ranging from 1 step/sec to 16384 steps/sec. This corresponds to a volume change rate of about 5.4 x 10^{-4} to about 8.8 cc/sec on a 250 cc unit and to a volume change rate of about 1.7 x 10^{-3} to about 28 cc/sec on a 750 cc unit.

CAUTION: **Either the Supply Valve or the Output Valve must be open before the Jog option is used. A warning message will be given on the LCD screen if both are closed. Use the Control Valves option on the Position menu to open at least one of them.**

Keys on the keypad are used to select the desired direction and speed. A summary description of what keys are used and how they are used is given below. Following the description, there is an example showing how a specific motor speed can be set.

- **Up arrow key:** Starts emptying the cylinder; incrementally increases the volume change rate if the cylinder is emptying; incrementally decreases the volume change rate if the cylinder is filling.

- **Down arrow key:** Starts filling the cylinder; incrementally increases the volume change rate if the cylinder is filling; incrementally decreases the volume change rate if the cylinder is emptying.

- **0, 1, 2, 3, or 4 key:** Sets the power of ten which determines the steps/sec increment added or subtracted before an arrow key is pressed. (0 sets an increment of 1, 1 sets an increment of 10, 2 sets and increment of 100, etc.)

- **Alt key:** Sets the condition of maximum speed before an arrow key is pressed.

- **Esc key:** Stops the piston motion.

***NOTE:* If an arrow key is pressed after the Jog menu is selected and before a number key is pressed, the pump piston motor will start running at 10 steps/sec.**

The following example describes how to move the flow pump piston at a motor speed of 2435 steps/sec in a direction to empty the cylinder.

1. Select the Jog option from the Position menu.

2. Be sure the Output Valve is open.

3. Press the 3 key, then the up arrow key. The piston will begin to move at a motor speed of 1000 (10^3) steps/sec.

4. Press the up arrow key again and the speed will increase to 2000 steps/sec.

5. Now press the 2 key, then the up arrow key. The speed will now be 2100 steps/sec (an increase of 10^2 steps/sec).

6. Press the up arrow key three more times to set the speed at 2400 steps/sec.

7. Next, press the 1 key, then press the up arrow key three times to attain 2430 steps/sec.

8. Finally, press the 0 key and then press the up arrow key five times.

If an arrow key is pressed sequentially to decrease the piston speed until it stops, the next press will start the piston moving in the opposite direction.

The flow pump piston can be moved at the maximum motor speed without having to return to the Position menu by pressing the Alt key and then the up or down arrow key.

2.1.2.5. Control Valves

Selecting the Control Valves option on the Position menu will open the menu in Figure 10.

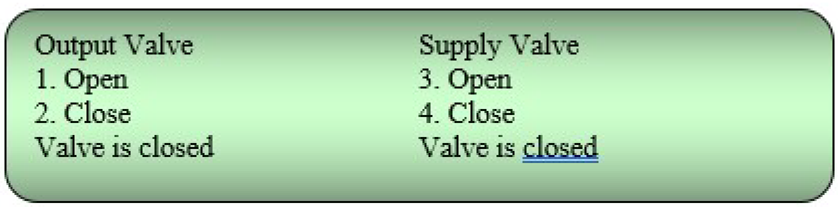


Figure 10. Control Valves Menu

As indicated on this menu, each valve can be opened or closed by pressing the appropriate numeric key on the keypad. When the valve is open, the front panel yellow LED for that valve will glow. The status of the valve is shown on the LCD screen.

2.1.3. Control

With the main menu displayed on the LCD screen, select the Control option by pressing 3 on the keypad. The Control menu (Figure 11) will appear on the LCD screen.

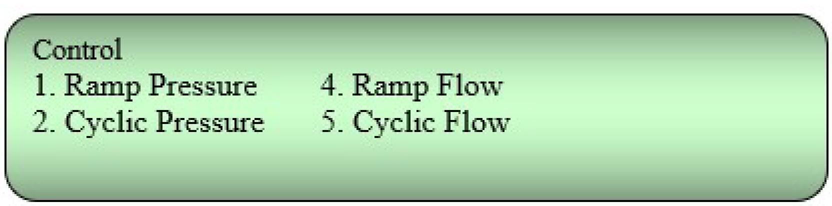


Figure 11. Control Menu

The options on this menu provide various ways pressure (i.e., load) or flow (i.e., strain) can be applied. While the change is being applied, the load cell and displacement transducer outputs are being read and the values stored in the controller memory at a rate determined by the sampling period you have chosen.

2.1.3.1. Ramp Pressure

With the Control menu displayed, press 1 to produce the menu in Figure 12.

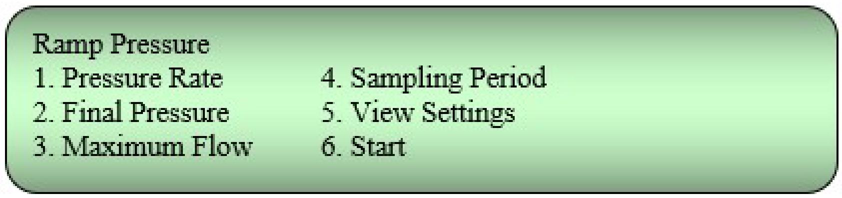


Figure 12. Ramp Pressure Menu

The first four options are used for entering numeric information. A value is entered by using the numeric keys and then pressing the Ent key. Pressing the left-pointing arrow will erase the digit to the left of the blinking cursor.

The Pressure Rate option shows the current rate at which a pressure change will occur and allows you to enter a new value for the rate. A value with no sign increases the pressure; a value preceded by a negative sign decreases the pressure.

The Final Pressure and Maximum Flow options show the current values for the pressure and flow that are related to the completion of a test and allow you to enter new values for these parameters. For ramp pressurizing, control will automatically stop and no more data will be taken only when the specified maximum flow is reached. If the specified final pressure is reached before the maximum flow, the pressure will be maintained and data will continue to be taken. It will be necessary for you to manually stop the test.

The Sampling Period option shows the current value for how often the sensors are read and data stored in the controller memory and allows you to enter a new value.

You can review the values entered in the first four options by selecting the View Settings option.

To start Ramp Pressure control, select the Start option and then press Ent. Readings of the load cell and displacement transducer (updated once a second) are shown on the display produced when you select the Start option. Also shown is the time (in seconds) which has elapsed from the start of the control. Control can be stopped by pressing Esc.

2.1.3.2. Cyclic Pressure

With the Control menu displayed, press 2 to produce the menu in Figure 13.

```
Cyclic Pressure
1. Seating Press        4. Sampling Period
2. Press Amplitude      5. View Settings
3. Cycle Period         6. Start
```

Figure 13. Cyclic Pressure Menu

The first four options are used for entering numeric information. A value is entered by using the numeric keys and then pressing the Ent key. Pressing the left-pointing arrow will erase the digit to the left of the blinking cursor.

The Seating Press option shows the current seating pressure value and allows you to enter a new value. The Press Amplitude option shows the current pressure amplitude value and allows you to enter a new value. The Cycle Period option shows the current cycle period value and allows you to enter a new value. The Sampling Period option shows the current value for how often the sensors are read and data stored in the controller memory and allows you to enter a new value.

You can review the values entered in the first four options by selecting the View Settings option.

To start Cyclic Pressure control, select the Start option and then press Ent. Current values for pressure and volume (updated once a second) are shown on the display produced when you select the Start option. Also shown is the time (in seconds) which has elapsed from the start of the control. Control can be stopped by pressing Esc.

2.1.3.3. Ramp Flow

With the Control menu displayed, press 3 to produce the menu in Figure 14.

```
Ramp Flow
1. Flow Rate            4. Sampling Period
2. Final Flow           5. View Settings
3. Maximum Press        6. Start
```

Figure 14. Ramp Flow Menu

The first four options are used for entering numeric information. Once you choose an option, you enter a value by using the number keys and then pressing the Ent key. Pressing the left-pointing arrow will erase the digit to the left of the blinking cursor.

The Flow Rate option shows the current rate at which a flow change will occur and allows you to enter a new value for the rate. A value with no sign increases the flow; a value preceded by a negative sign decreases the flow.

The Final Flow and Maximum Press options show the current values for the flow and pressure that are related to the completion of a test and allow you to enter new values for these parameters. For ramp flow, control will automatically stop and no more data will be taken only when the specified maximum pressure is reached. If the specified final flow is reached before the maximum pressure, the flow will be maintained and data will continue to be taken. It will be necessary for you to manually stop the test.

The Sampling Period option shows the current sampling period and allows you to enter a new sampling period.

You can review the values entered in the first four options by selecting the View Settings option.

To start Ramp Flow control, select the Start option and then press Ent. Readings for flow and pressure (updated once a second) are shown on the display produced when you select the Start option. Also shown is the time (in seconds) which has elapsed from the start of the control. Control can be stopped by pressing Esc.

2.1.3.4. Cyclic Flow

With the Control menu displayed, press 4 to produce the menu in Figure 15.

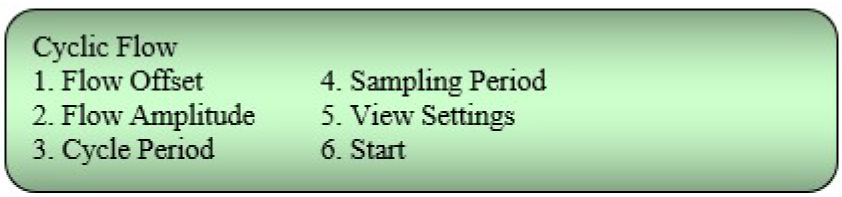


Figure 15. Cyclic Flow Menu

The first four options are used for entering numeric information. Once you choose an option, you enter a value by using the number keys and then pressing the Ent key. Pressing the left-pointing arrow will erase the digit to the left of the blinking cursor.

The Flow Offset option shows the current flow offset and allows you to enter a new flow offset. The Flow Amplitude option shows the current flow amplitude and allows you to enter a new flow amplitude. The Cycle Period option shows the current cycle period and allows you to enter a new cycle period. The Sampling Period option shows the current sampling period and allows you to enter a new sampling period.

You can review the values entered in the first four options by selecting the View Settings option. Then you can start the load change by selecting the Start option and pressing Ent. Readings of the load cell and displacement transducer (updated once a second) are shown on the display produced when you select the Start option. Also shown is the time (in seconds) which has elapsed from the start of the control. Control can be stopped by pressing Esc.

2.1.4. Setup

With the main menu displayed on the LCD screen, select the Setup option by pressing keypad 4. The Setup menu (Figure 16) will appear on the LCD screen.

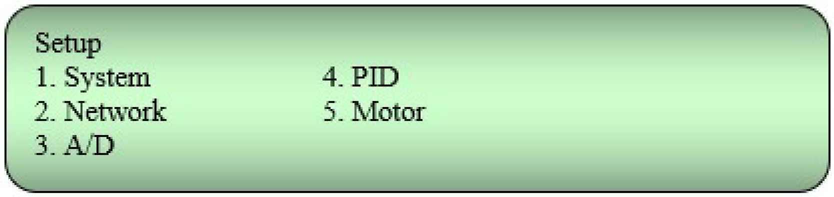


Figure 16. Setup Menu

The information entered by way of the various options on the Setup menu is necessary for the FlowTrac unit to function correctly. All information is entered by way of the keypad. In addition, calibration factor and offset values can be downloaded from the software being used to control a test involving one or more FlowTrac units.

Table 1 shows the default parameters that were loaded into the FlowTrac unit at the time it was shipped. Below the table are detailed descriptions of the menus accessed from the Setup menu and of the parameters entered by way of these menus.

<table>
<tr><th>MENU</th><th>ITEM</th><th colspan="20">PARAMETERS</th></tr>
<tr><td rowspan="2">1. System</td><td>1. A/D Channel</td><td colspan="20">1 or 2 depending on intended FlowTrac use</td></tr>
<tr><td>2. Units</td><td colspan="20">Metric (or English if you prefer)</td></tr>
<tr><td>2. Network</td><td>1. Node ID</td><td colspan="20">66 (cell pressure unit), 67 (sample or bottom pressure unit), 68 (top pressure unit) (Can be changed to any value between 20 and 256.)</td></tr>
<tr><td rowspan="16">3. A/D</td><td rowspan="8">1. Channel 1</td><td colspan="5">1. Type</td><td colspan="5">2. Excitation</td><td colspan="5">3. Polarity</td><td colspan="5">4. Range</td></tr>
<tr><td colspan="5">Pressure</td><td colspan="5">12 V</td><td colspan="5">Bipolar</td><td colspan="5">160 mV</td></tr>
<tr><td colspan="5">5. Factor*</td><td colspan="5">6. Offset</td><td colspan="5">7. Low. Lim.</td><td colspan="5">8. Up. Lim.</td></tr>
<tr><td colspan="5">$5.6x10^{-2}$ kPa/cnt</td><td colspan="5">32768 cnt</td><td colspan="5">0 cnt</td><td colspan="5">65535 cnt</td></tr>
<tr><td colspan="20">9. Advanced</td></tr>
<tr><td colspan="4">1. Mode</td><td colspan="4">2. Rate</td><td colspan="4">3. Filter</td><td colspan="4">4. Chop</td><td colspan="4">5. Fast</td></tr>
<tr><td colspan="4">Diff.</td><td colspan="4">200 HZ</td><td colspan="4">Enabled</td><td colspan="4">Enabled</td><td colspan="4">Enabled</td></tr>
<tr></tr>
<tr><td rowspan="8">2. Channel 2**
(Triaxial test parameters)</td><td colspan="5">1. Type</td><td colspan="5">2. Excitation</td><td colspan="5">3. Polarity</td><td colspan="5">4. Range</td></tr>
<tr><td colspan="5">Pressure</td><td colspan="5">12 V</td><td colspan="5">Bipolar</td><td colspan="5">160 mV</td></tr>
<tr><td colspan="5">5. Factor*</td><td colspan="5">6. Offset</td><td colspan="5">7. Low. Lim.</td><td colspan="5">8. Up. Lim.</td></tr>
<tr><td colspan="5">$5.6x10^{-2}$ kPa/cnt</td><td colspan="5">32768 cnt</td><td colspan="5">0 cnt</td><td colspan="5">65535 cnt</td></tr>
<tr><td colspan="20">9. Advanced</td></tr>
<tr><td colspan="4">1. Mode</td><td colspan="4">2. Rate</td><td colspan="4">3. Filter</td><td colspan="4">4. Chop</td><td colspan="4">5. Fast</td></tr>
<tr><td colspan="4">Diff.</td><td colspan="4">200 Hz</td><td colspan="4">Enabled</td><td colspan="4">Enabled</td><td colspan="4">Enabled</td></tr>
<tr></tr>
<tr><td rowspan="7">4. PID</td><td colspan="21">
<table>
<tr><td>1. P-Gain</td><td>2. I-Gain</td><td>3. D-Gain</td><td>4. V-Offset</td></tr>
<tr><td>3 steps/sec/cnt</td><td>2 steps/sec/cnt</td><td>0 steps/sec/cnt</td><td>0 steps/sec</td></tr>
<tr><td>5. I-Limit</td><td>6. V-Limit</td><td>7. Dither Ampl.</td><td>8. Dither Freq.</td></tr>
<tr><td>1024 steps/sec</td><td>32768 steps/sec</td><td>0 steps/sec</td><td>0 Hz</td></tr>
<tr><td colspan="4">9. Advanced</td></tr>
<tr><td>1. Filter</td><td>2. FB Channel</td><td>3. Anti-Windup</td><td>4. Derivative FB</td></tr>
<tr><td>256 Hz</td><td>Channel 0</td><td>Reset</td><td>De/dt</td></tr>
</table>
</td></tr>
<tr></tr><tr></tr><tr></tr><tr></tr><tr></tr><tr></tr>
<tr><td rowspan="4">5. Motor</td><td>1. Frequency</td><td colspan="20">16384 Hz</td></tr>
<tr><td>2. Acceleration</td><td colspan="20">40960 steps/sec/sec</td></tr>
<tr><td>3. Step Capacity</td><td colspan="20">416213</td></tr>
<tr><td>4. Step factor</td><td colspan="20">$5.4x10^{-4}$ cc/step, 250 cc unit; $1.7x10^{-3}$ cc/step, 750 cc unit</td></tr>
</table>

*The factor shown is approximate. It is unique to each sensor and needs to be updated after each calibration procedure.

**The settings shown for Channel 2 are those for when a FlowTrac unit is used for Triaxial tests. If the unit will be used for the cell pressure in Rowe Consolidation tests, Channel 2 will be set up for a displacement transducer. See Table 2 below for setup parameters.

Table 1. Summary of FlowTrac Settings

<table>
<tr><th colspan="22">Channel 2 Configuration for Rowe Consolidation Tests</th></tr>
<tr><th>FlowTrac Use</th><th>Channel 2 use</th><th colspan="20">Settings</th></tr>
<tr><td rowspan="8">Cell pressure</td><td rowspan="8">Vertical Displacement</td><td colspan="5">1. Type</td><td colspan="5">2. Excitation</td><td colspan="5">3. Polarity</td><td colspan="5">4. Range</td></tr>
<tr><td colspan="5">Displacement</td><td colspan="5">5 V</td><td colspan="5">Unipolar</td><td colspan="5">2560</td></tr>
<tr><td colspan="5">5. Factor*</td><td colspan="5">6. Offset</td><td colspan="5">7. Low. Lim.</td><td colspan="5">8. Up. Lim.</td></tr>
<tr><td colspan="5">8.0×10^{-4} mm/cnt</td><td colspan="5">32768 cnt</td><td colspan="5">0 cnt</td><td colspan="5">65535 cnt</td></tr>
<tr><td colspan="20">9. Advanced</td></tr>
<tr><td colspan="4">1. Mode</td><td colspan="4">2. Rate</td><td colspan="4">3. Filter</td><td colspan="4">4. Chop</td><td colspan="4">5. Fast</td></tr>
<tr><td colspan="4">Diff.</td><td colspan="4">200 Hz</td><td colspan="4">Enabled</td><td colspan="4">Enabled</td><td colspan="4">Enabled</td></tr>
<tr></tr>
<tr><td rowspan="7">Sample pressure</td><td rowspan="7">External Sample Pressure</td><td colspan="5">1. Type</td><td colspan="5">2. Excitation</td><td colspan="5">3. Polarity</td><td colspan="5">4. Range</td></tr>
<tr><td colspan="5">Pressure</td><td colspan="5">12 V</td><td colspan="5">Bipolar</td><td colspan="5">160 mV</td></tr>
<tr><td colspan="5">5. Factor*</td><td colspan="5">6. Offset</td><td colspan="5">7. Low. Lim.</td><td colspan="5">8. Up. Lim.</td></tr>
<tr><td colspan="5">5.6×10^{-2} kPa/cnt</td><td colspan="5">32768 cnt</td><td colspan="5">0 cnt</td><td colspan="5">65535 cnt</td></tr>
<tr><td colspan="20">9. Advanced</td></tr>
<tr><td colspan="4">1. Mode</td><td colspan="4">2. Rate</td><td colspan="4">3. Filter</td><td colspan="4">4. Chop</td><td colspan="4">5. Fast</td></tr>
<tr><td colspan="4">Diff.</td><td colspan="4">200 Hz</td><td colspan="4">Enabled</td><td colspan="4">Enabled</td><td colspan="4">Enabled</td></tr>
</table>

*The factor shown is approximate. It is unique to each sensor and needs to be updated after each calibration procedure.

Table 2. Summary of Rowe Consolidation Test Settings

2.1.4.1. System

With the Setup menu displayed on the LCD screen, select the System option. The System Setup menu (Figure 17) will appear on the LCD screen.

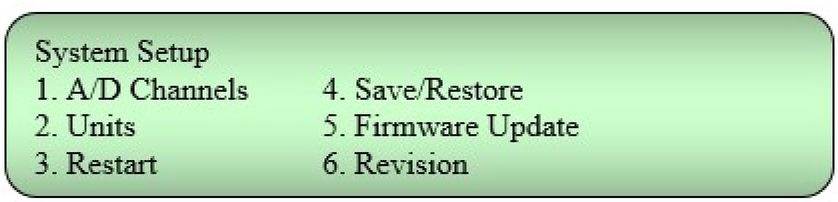


Figure 17. System Setup Menu

The A/D Channels option, which brings up the A/D Channel Count Setup entry screen, is for specifying how many channels (signal paths) will be processed by the controller in the FlowTrac unit. The controller is capable of processing four channels. However, only two channels (channels 1 and 2) can have sensors connected to them. Channel 1 is for the internal pressure sensor. A second sensor can be attached to the connector on the back panel and would be processed by channel 2.

For the most common use of FlowTrac , only channel 1 is used. When the FlowTrac unit is used for cyclic triaxial or Rowe consolidation testing, both channels 1 and 2 are used. The default setting for your unit will depend on how your unit will be used. The default value shown on the A/D Channel Count Setup screen should not be changed. The number of channels shown there will determine what is shown on any LCD display with channel information.

If your unit has been set up for only one channel and you need to add an additional sensor, you will have to change the number on the A/D Channel Count Setup screen to 2 and set up that sensor using the A/D Setup menu (see Section 2.1.4.3 A/D (Analog to Digital)).

The Units option, which brings up the Units Setup menu, is for choosing whether metric or English units will be used for displays which require engineering units. When a different set of units is chosen, all displays are automatically changed to show the value correctly in the units chosen. There is no need to turn off FlowTrac and turn it back on (or to select the Restart option on the System Setup menu) for a change to become effective.

The Restart option, when selected by pressing 3, has the same effect as turning off the load frame and turning it back on again.

The Save/Restore option brings up the System Setup Save/Restore menu (Figure 18) which allows you to save or restore setup parameters. There are two reserved sections of controller board memory, designated by A and B, in which test parameters currently being used can be saved. Parameters saved in either section can be restored to the active portion of memory (i.e., that portion of memory used by the load frame when a test is run manually).

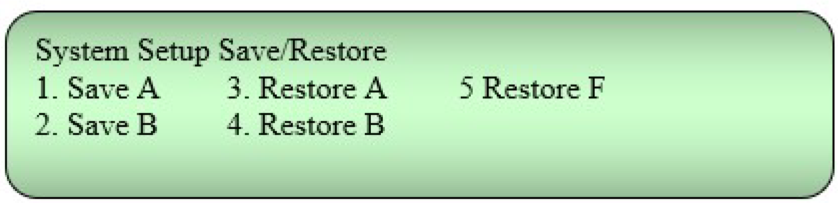


Figure 18. System Setup Save/Restore Menu

Before a FlowTrac unit is shipped to you, setup parameters specific to your system (calibration factors, offsets, parameters specific to the type of test for which you ordered your system, etc.) are loaded into the active portion of the control board memory. If you plan to run a different type of test and want to save the parameters for the original test type, you can do so by selecting either Save A or Save B. Then you can modify test parameters to be what are needed for the new type of test.

If you want to run the original type test again, you can save the new parameters in the other reserved section of memory and select the restore option corresponding to the saved original parameters.

Example

You ordered a system for doing triaxial tests and have done several of them. You now want to do Rowe consolidation tests. You know that after running several Rowe consolidation tests, you will want to run some triaxial tests again. The following steps lead you through what you need to do.

1. With the System Setup Save/Restore menu showing, press 1 on the keyboard to save the triaxial parameters in memory section A. ("System setup A saved" will appear at the bottom of the LCD screen.)

2. Change setup parameters to those for a Rowe consolidation test. (You will probably need to change sensor offsets and motor parameters, and possibly PID settings.)

3. With the System Setup Save/Restore menu showing, press 2 on the keyboard to save the Rowe consolidation parameters in memory section B.

4. After you have run the Rowe consolidation tests, display the System Setup Save/Restore menu and press 3 on the keyboard. ("System setup A restored" will appear at the bottom of the LCD screen.) This will restore the triaxial parameters to the active memory.

5. Reboot the load frame (or select the Restart option from the System Setup menu).

6. Now if you want to run the Rowe consolidation tests again, display the System Setup Save/Restore menu and press 4. (After restoring parameters, make sure to reboot the load frame or select the Restart option from the System Setup menu.)

During the fabrication of a FlowTrac unit, a basic, generic set of parameters are loaded into a portion of the control board memory designated by F (for Factory). On the very rare occasion that the active portion of memory becomes corrupted and the board cannot complete its normal boot process, these parameters will be used. This allows for the use of the debug tools built into the board. If you notice that the setup parameters have changed from what you last put in, call Geocomp, Inc. for assistance.

The Firmware Update option shows the firmware version that is currently loaded into the board and has a line for showing an available revision. Normally this line will read N/A (Not Available). Should it happen that you have acquired from Geocomp, Inc. a recent revision and have the HEX file for that version on the hard drive of your computer, you can use the DIAGS program to upload the revision to the control board (see your DIAGS software manual for details). When this is done, the new version will be shown on the Available Revision line. If an available revision is shown, you can make it the current revision by, while the Firmware Update screen is showing, pressing the Alt key and then the up arrow key on the keypad. When you do this, the new version will be shown on the first line and remain showing on the Available Revision line.

The Revision option displays the current version of the firmware and the hardware. It also shows the size of memory (called the flash memory) in which setup parameters are stored and which holds the data taken during a test. The flash memory is capable of holding about 64,000 data points.

2.1.4.2. Network

With the Setup menu displayed on the LCD screen, select the Network option. The Node ID Setup menu (Figure 19) will appear on the LCD.

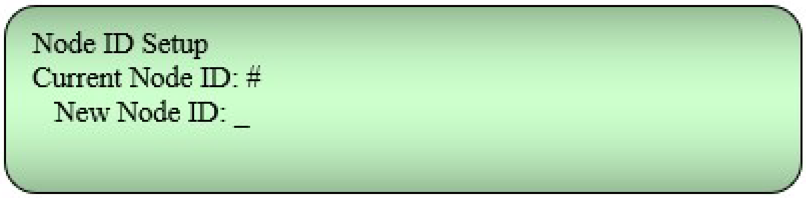


Figure 19. Node ID Setup Screen

The Node ID Setup screen is used to specify the identification number needed by the software to determine which FlowTrac unit it will control. If a test system has only one FlowTrac unit, the default value is 66. Additional units have default Node ID numbers of 67 and 68. When two units are used in a ROWE consolidation, a triaxial or a stress path test, the Node IDs should be 66 and 67 for cell pressure control and sample pressure control, respectively. When three units are used in a permeability test, the Node IDs should be 66, 67, and 68 for cell pressure, bottom pressure, and top pressure, respectively.

If you will be controlling more than one test system with your computer, you will need to change the Node ID numbers for units in the other systems from the default values given above. Any number between 10 and 256 can be used. If you change a Node ID, you will need to turn off the FlowTrac unit and turn it on again (or select the Restart option on the System Setup menu) for the change to become effective.

> *CAUTION: **The Node ID on the LCD entry screen and the Node ID on the software Hardware Setup window must be the same. Do not change an LCD entry screen Node ID unless you also change the software Node ID. It is best if you consult a Geocomp, Inc. technician before making any changes.**

2.1.4.3. A/D (Analog to Digital)

With the Setup menu displayed on the LCD screen, select the A/D option. The A/D Setup menu (Figure 20) will appear on the LCD screen. If your FlowTrac unit will use only its internal pressure transducer, the screen will list only Channel 1 as shown below.

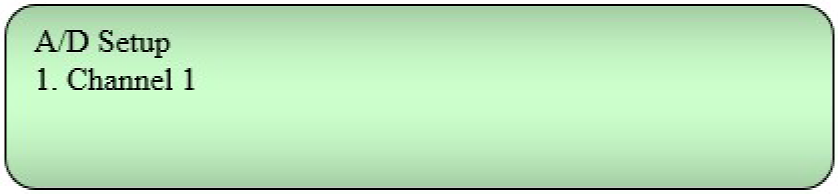


Figure 20. A/D Setup Menu

If your FlowTrac unit will be used for Rowe consolidation or cyclic triaxial tests, the screen should list both Channel 1 and Channel 2.

Note that the number of channels listed depends on the number entered on the A/D Channel Count Setup screen (accessed from the System Setup menu, option 1).

When a channel is selected from the A/D Setup menu, the A/D Channel X Setup menu (Figure 21) will be displayed as shown below, where X stands for 1 or 2.

Figure 21. A/D Channel Setup Menu

*CAUTION: **All settings were specified at the factory before the FlowTrac unit was shipped. Except for the Factor and Offset, there should be no need to change any of the A/D setup parameters. If there is reason to believe a setting should be changed, consult a technician at Geocomp, Inc. before making the change.**

The Type option is for assigning a particular type of transducer to the channel selected. The eight types that can be used are: Generic, Force, Pressure, Volume, Displacement, Temperature, Rotation and Torque. The Generic choice is used only when the transducer is not one of the seven types in the list.

The current type assigned to the selected channel is shown on the display produced when the Type option is selected. For all uses except cell pressure in Rowe consolidation tests, Pressure would be assigned to both Channels 1 and 2. (For cell pressure use in Rowe consolidation tests, Displacement is assigned to Channel 2.) If you change the sensor type for Channel 2 and configure that sensor using the A/D Channel 2 Setup menu, you will need to turn off the FlowTrac unit and turn it on again (or select the Restart option on the System Setup menu) for the change to become effective.

The Excitation option is used for setting the excitation voltage that will be applied to the sensor connected to the selected channel. When the Excitation option is chosen, the current excitation for the channel is shown and the option given to enter a new excitation. A value is entered by using the numeric keys and then pressing the Ent key. Pressing the left-pointing arrow will erase the digit to the left of the blinking cursor.

The Polarity option is for specifying whether the sensor is unipolar (always positive relative to ground) or bipolar (either positive or negative relative to ground). A sensor can also be set as Bipolar/Reversed or Unipolar/Reversed. If a reversed polarity is selected, an increasing output from the sensor results in decreasing values recorded by the control board (and by controlling software when being used), and vice versa.

When the Polarity option is chosen, the current polarity for the channel is shown and the option given to select a different polarity. The FlowTrac pressure transducer (Channel 1) should be set as bipolar. If you assign a polarity to a sensor you have added, you will need to turn off the FlowTrac unit and turn it on again (or select the Restart option on the System Setup menu) for the change to become effective.

The Range option is for assigning a range of voltages that the selected channel can process. The ranges available (in mV) are 2560 (i.e., 0 – 2560), 1280, 640, 320, 160, 80, 40 and 20. When the Range option is chosen, these ranges along with the current range for the channel are shown, and the option is given to select a different range. The value selected needs to be equal to or larger than the full scale output of the sensor.

Example: If the specifications for a sensor show (10 mV output)/(Volt excitation) at full scale, and the sensor is powered with a 12-Volt excitation, then the full-scale output would be 10 mV/V x 12 V = 120 mV. For this sensor, the Range should be set at 160.

The range that is assigned to the internal pressure transducer should be 160. If you assign a range to a transducer you have added, you will need to turn off the FlowTrac unit (or select the Restart option on the System Setup menu) and turn it on again (or select the Restart option on the System Setup menu) for the change to become effective.

The Factor and Offset options are used to enter calibration parameters. The calibration factor and offset convert the transducer readings from counts to engineering units according to the equation,

$SR_{EU} = CF\left(SR_{cnts} - O_{cnts}\right)$ in which

SR_{EU} = Sensor Reading in Engineering Units,
CF = Calibration Factor,
SR_{cnts} = Sensor Reading in Counts and
O_{cnts} = Offset in Counts.

The embedded firmware performs the conversion by first subtracting the offset (in counts) from the transducer reading (in counts) and then multiplying the result by the conversion factor. The pressure offset corresponds to the appropriate pressure transducer reading at the beginning of a test. (The condition for the reading will be different for different types of tests. See your software manual for details.)

The calibration factor value will depend on the system of units you are using. The value will be automatically adjusted if a new system of units is selected. The current units are shown on the System Monitor window.

If the FlowTrac unit is being used as a stand-alone unit under manual control, each calibration factor and offset value must be entered using the keypad and the Ent key. After entering a new value, it is not necessary to turn off the FlowTrac unit and turn it on again (or select the Restart option on the System Setup menu). If the flow pump unit is being used for automated tests under the control of a computer, the calibration and offset values can be either entered manually or downloaded from the control program being used to run the test. (See your software manual for calibration procedures and downloading of results.)

The Lower Limit and Upper Limit options are used to set minimum and maximum transducer outputs (in counts). The transducer readings are checked continuously when the FlowTrac unit is under manual control. If the controller finds that a reading is outside one of these limits, the unit stops functioning until the condition the transducer is measuring has been adjusted to be within limits. The lower and upper limits are set to 0 and 65535 respectively and should not be changed.

When the Advanced option is selected, the A/D Advanced Channel X Setup menu (Figure 22) will be displayed.

Figure 22. A/D Advanced Channel X Setup Menu

This menu is used to set data reading parameters. The Mode can be Differential or Single Ended. The Rate can be set at one of eight different values ranging from 200 Hz to 1024 Hz. The Filter can be disabled or enabled. The Chop condition can be disabled or enabled. Fast data reading can be disabled or enabled. The default conditions are set at the factory should not be changed.

2.1.4.4. PID

With the Setup menu displayed on the LCD screen, select the PID option. The PID Setup menu (Figure 23) will appear on the LCD screen.

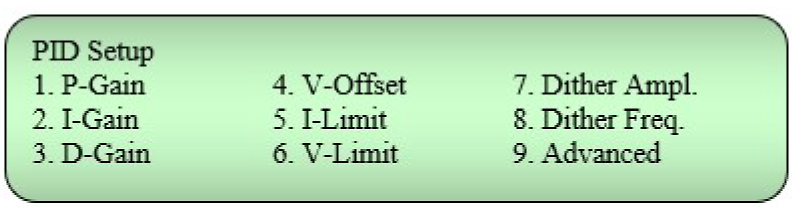


Figure 23. PID Setup Menu

The options on this menu are used to set the parameters that govern how the flow pump motor will run. The first three options of Proportional Gain, Integral Gain and Derivative Gain are parameters that govern the motor behavior when a target value for pressure or volume has been set for a test. The sixth option, Velocity Limit, sets the speed the motor will run when the Position option of Empty or Fill is chosen. It also sets the limit on how fast the Jog option can run the motor.

When the Advanced option is selected, the PID Advanced Setup menu (Figure 24) will be displayed.

Figure 24. PID Advanced Setup Menu

The Advanced option is used to set various advanced, seldom changed feedback loop parameters. The PID Advanced Setup default parameters have been set at the factory for optimal performance and should not be changed.

CAUTION: **It is very important that you consult with Geocomp, Inc. before making any changes to the PID parameters other than P-Gain and I-Gain.**

2.1.4.5. Motor

With the Setup menu displayed on the LCD screen, select the Motor option. The Motor Setup (Figure 25) menu will appear on the LCD screen.

Figure 25. Motor Setup Menu

The Frequency option is used to set the maximum step rate at which the pump motor can be run. The values start at 512 and increase by multiples of two to 65536 steps/sec (designated on the LCD display as Hz). The frequency setting determines the minimum time that can occur between each step. A smaller frequency setting corresponds to a longer minimum time between steps. The default frequency setting of 16384 steps/sec should be lowered only if it is necessary to have a minimum period longer than 0.1 ms between each step. If you change the setting, you will need to turn off the FlowTrac unit and turn it on again (or select the Restart option on the System Setup menu) for the change to become effective.

The Acceleration option is used to set the rate at which the motor will speed up or slow down. The default value of 40960 steps/sec^2 is set at the factory for optimal performance, and should not be changed.

The Step Capacity option is used to specify the number of steps the motor goes through as it moves the pump piston from the Full limit switch to the Empty limit switch. The value is unique to each flow pump and is set at the factory. It should not be changed. When the pump cylinder is being emptied using the Empty option from the Position menu, the Step Count will be set to this value when the Empty limit switch is activated. If there is reason to believe the default value has been changed, consult a technician at Geocomp, Inc. for resetting instructions.

The Step Factor option is used to set the Step Factor. This factor is used to convert steps to volume change (i.e., it is the flow pump cylinder volume change that occurs for one step). The Step Factor depends on the physical construction of the FlowTrac unit and has been set at the factory. It should not be changed. If there is reason to believe the default value has been changed, consult a technician at Geocomp, Inc. for resetting instructions.

2.2. Computer Controlled Operation

Under computer control, the FlowTrac system is capable of performing a variety of automated tests. Rowe incremental consolidation tests and semi-automated permeability (constant gradient or constant flow) tests can be run using two FlowTrac units. Fully-automated permeability (constant gradient or constant flow), triaxial (all types), stress path and CRC tests can be run with the addition of a LoadTrac-II load frame. The following descriptions explain how the hardware is configured for several types of tests. Refer to your software manual for an explanation of the control program which will automatically run and report the test.

2.2.1. CRC Test

The automation of a Constant Rate of Consolidation (CRC) test requires a LoadTrac-II load frame, one FlowTrac unit, a CRC cell and a computer. (The additional equipment is available from Geocomp, Inc.) Figure 26 shows the equipment and the hydraulic connections that are needed.

Figure 26. LoadTrac-II/FlowTrac System for Constant Rate of Consolidation (CRC) Tests, Fully Automated

The software package that fully automates the CRC test is available from Geocomp, Inc. The software user's manual explains the specimen preparation, the positioning of the CRC cell on the load frame platen, the initialization of the flow pump, the running of the test and the form of the report generated.

2.2.2. Permeability (Hydraulic Conductivity) Test, Fully Automated

The full automation of a permeability test requires one LoadTrac-II unit, three FlowTrac units, a flexible wall permeability cell and a computer. (The additional equipment is available from Geocomp, Inc.) Figure 27 shows the equipment and the hydraulic connections that are needed.

Figure 27. LoadTrac-II/FlowTrac System for Permeability (Hydraulic Conductivity) Tests, Fully Automated

The software package that controls the complete automation of the test is called PERM and is available from Geocomp, Inc. The software user's manual explains the specimen preparation, the initialization of the flow pumps, the running of the test and the form of the report generated.

2.2.3. Permeability (Hydraulic Conductivity) Test, Semi-Automated

The partial automation of a permeability test requires two FlowTrac units, a flexible wall permeability cell, an external source of compressed air (1000 kPa) and a computer. (The additional equipment is available from Geocomp, Inc.) The FlowTrac units are used to automate the initialization, saturation and consolidation phases by controlling the cell and sample pressure and volume. Then the test is momentarily halted before the beginning of the flow phase so that manual changes in the line connections to the permeability cell can be made. Finally, the flow pumps automate the flow phase by controlling the top sample and bottom sample pressures while an external source of compressed air applies a constant cell pressure. Figure 28 shows the equipment the hydraulic connections that are needed.

Figure 28. FlowTrac System for Permeability (Hydraulic Conductivity) Tests, Semi-Automated

The software package that controls the automated portion of the test is called FLOW and is available from Geocomp, Inc. The software user's manual explains the specimen preparation, the initialization of the flow pumps, the running of the test and the form of the report generated.

2.2.4. Rowe Test

The automation of a Rowe incremental consolidation test requires two FlowTrac flow pumps, a Rowe cell, and a computer. (The additional equipment is available from Geocomp, Inc.) Figure 29 shows the equipment and hydraulic connections that are needed and Figure 30 shows the hydraulic and electrical connections.

Figure 29. FlowTrac System for Rowe Incremental Consolidation Tests

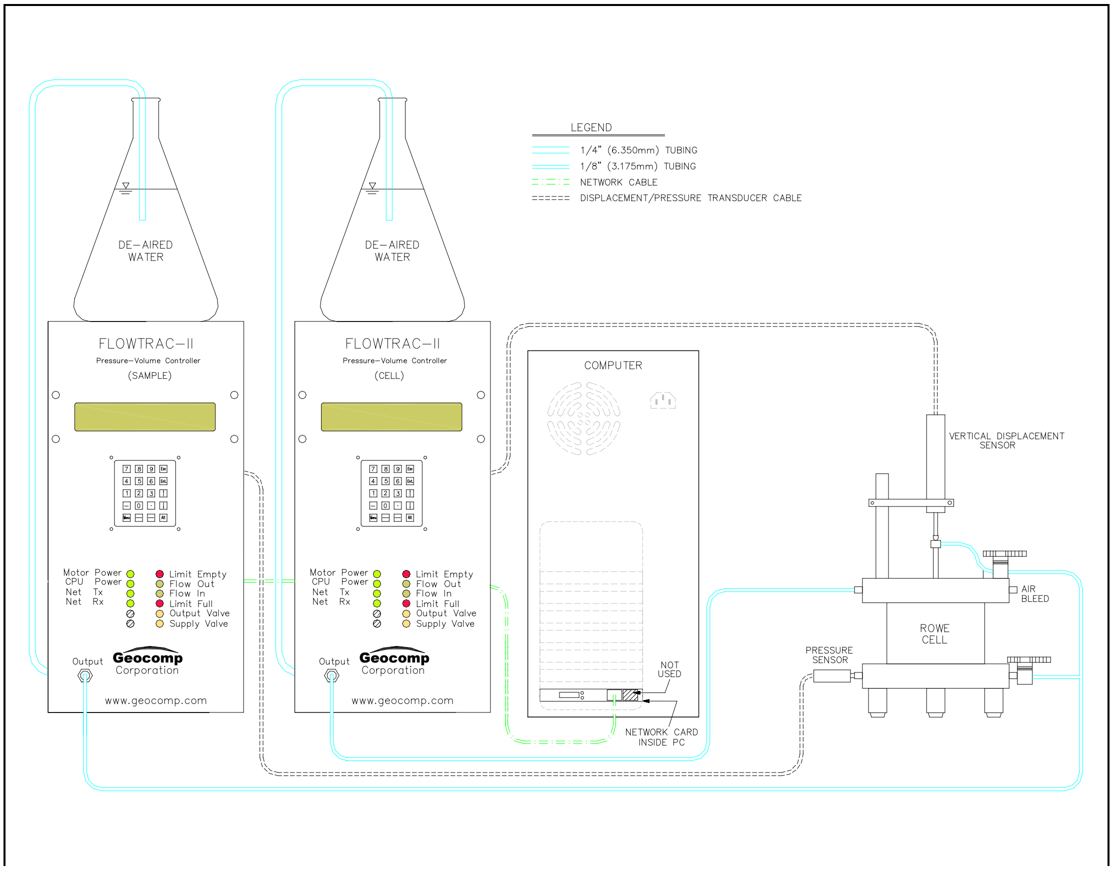


Figure 30. Diagram Showing FlowTrac System for Rowe Incremental Consolidation Testing

The software package that fully automates the Rowe test is called RCON and is available from Geocomp, Inc. The software user's manual explains the specimen preparation, the initialization of the flow pump the running of the test and the form of the report generated.

2.2.5. Triaxial and Stress Path Test

The automation of a triaxial and stress path test requires a LoadTrac-II load frame, two FlowTrac units, a triaxial cell and a computer. (The additional equipment is available from Geocomp, Inc.) Figure 31 shows the equipment that is needed. Figure 32 shows the hydraulic and electrical connections.

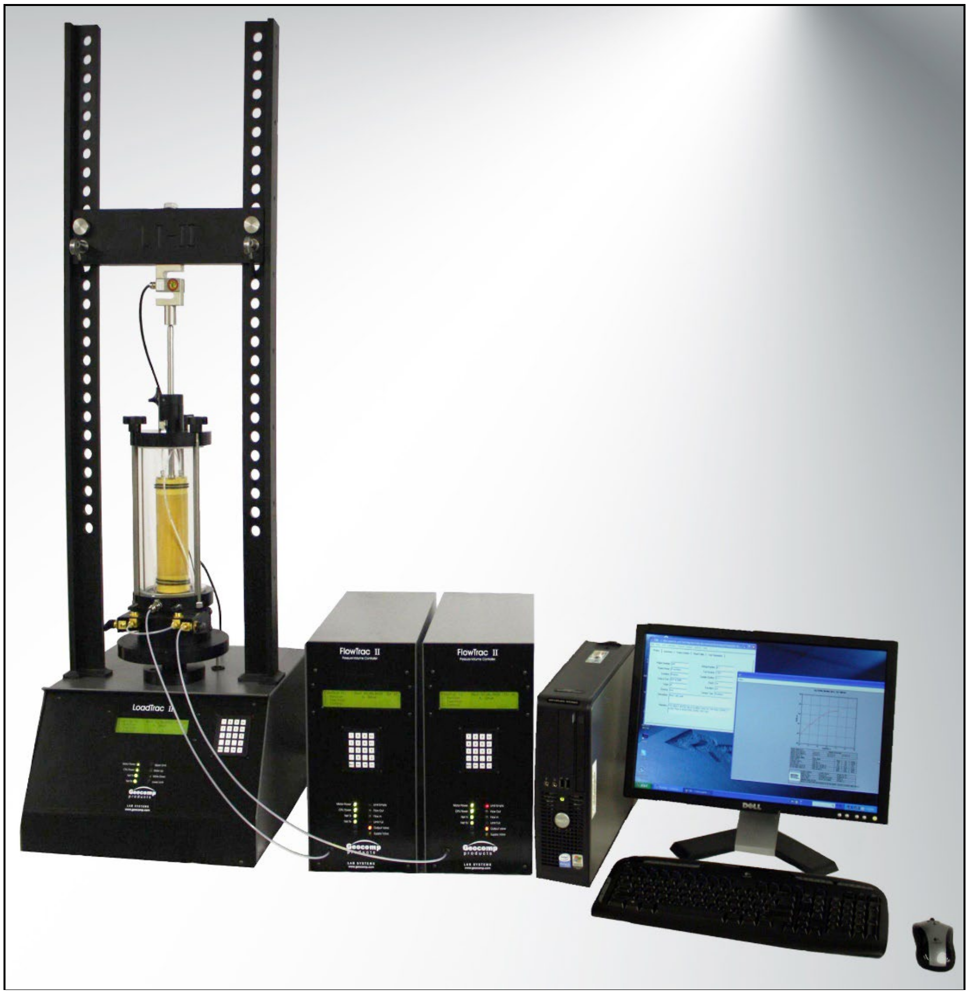
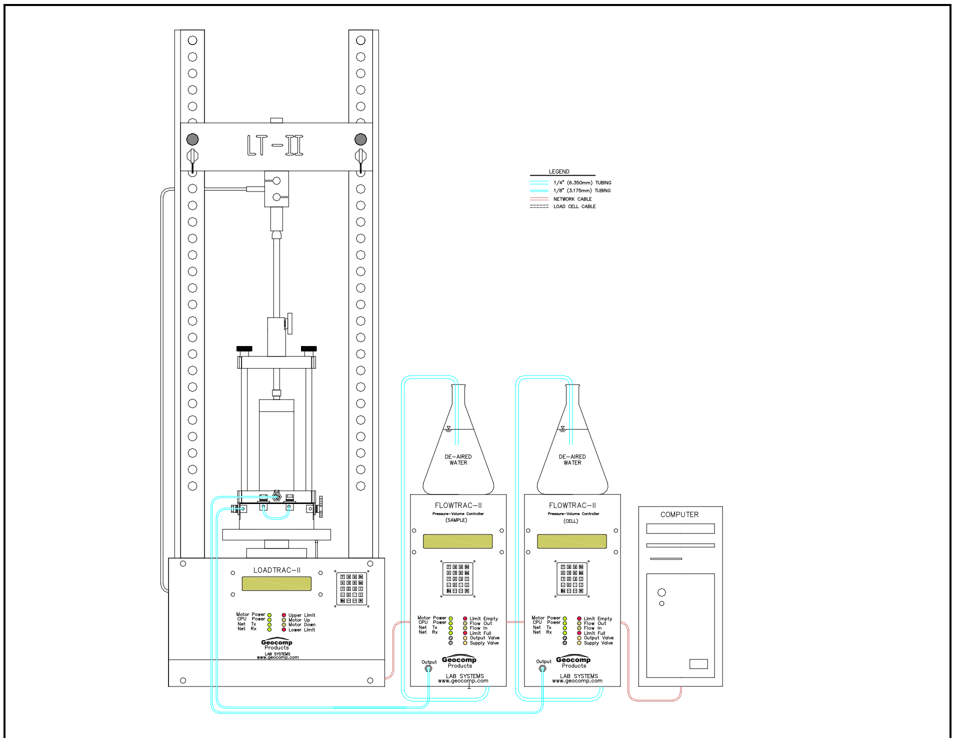
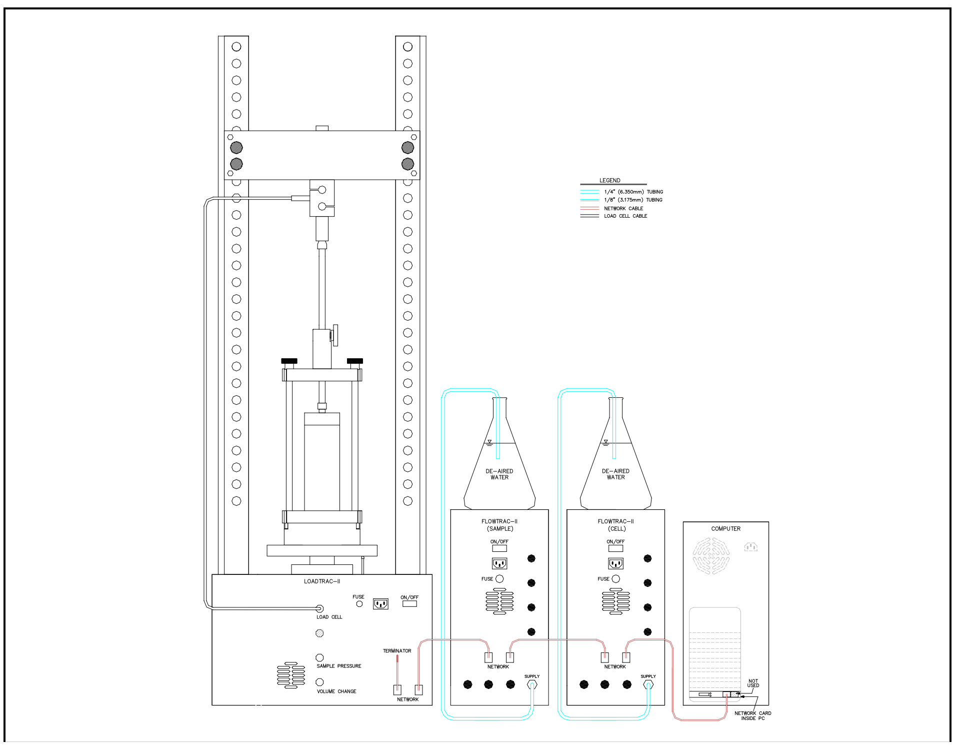


Figure 31. LoadTrac-II/FlowTrac System for Triaxial and Stress Path Tests, Fully Automated



(A) Front View



(B) Back View

Figure 32. Installation Diagrams for FlowTrac /LoadTrac-II System for Triaxial (all types) and Stress Path Tests

The software package that fully automates triaxial and stress path tests is called TRIAXIAL and is available from Geocomp, Inc. The software User's Manual explains the specimen preparation, the positioning of the triaxial cell on the load frame platen, the initialization of the flow pumps, the running of the test and the form of the report generated.

APPENDIX A: INSTALLATION

A.1. Preparing Test Station

Select a location for your FlowTrac system components that is far away from dust and dirt generators such as sieving and compacting equipment. (While the FlowTrac apparatus is reasonably insensitive to dust and dirt, excessive amounts may clog the keyboard, foul the CD drive, or otherwise inhibit the functioning of the equipment.)

The FlowTrac system (one FlowTrac unit, computer and monitor) requires three electrical outlets and draws a maximum total current of 2.5 A at 120 V AC. We strongly recommend that you use a circuit which is totally dedicated to the FlowTrac system in order to minimize electrical noise from other equipment in your lab. You may want to install a filter/surge protector or an Uninterruptible Power Source (UPS) between the wall outlet and your FlowTrac equipment. (A UPS provides excellent isolation, and also keeps the system running in the event of a power failure.) A UPS is particularly important if your lab experiences frequent power losses or brownouts. Contact Geocomp, Inc. for UPS information.

The FlowTrac system should be placed on a bench, desk, table or other work area with a surface approximately 0.8 m (30 in) above the floor. The work area support should be stable enough to prevent shaking if bumped by lab personnel, and be isolated from any other laboratory equipment that produces vibrations.

Each FlowTrac unit takes up a surface area approximately 0.2 m wide by 0.5 m deep (8 in by 20 in). The computer, monitor and keyboard take up a surface area of about 0.61 m wide by 0.66 m deep (24 in by 26 in). To save workspace, the computer can be placed on the floor under the work surface after the Network/Communication card has been installed.

Place your computer, keyboard and monitor on the work surface. Leave plenty of space around the computer because you will need to remove its case in order to install the Network/Communication card. Leave all cables disconnected from your computer.

A.2. Unpacking

Immediately upon receiving your FlowTrac system, examine the shipping crate(s) for indications of shipping damage. Notify Geocomp, Inc. at once if you suspect damage from shipping. Check the packing list(s) to ensure that you received the correct number of crates. Check all items in the crate(s) against the packing list(s) to ensure that you received everything.

<u>Accessory Items</u>

Locate the test accessories (test cells, etc.) and place them on the work surface. Locate the miscellaneous items (power cords, network cable(s), CD, etc.) and place them on the work surface. Locate the Network/Communication card and set it on the work surface next to your computer.

FlowTrac Unit(s)

Each FlowTrac unit weighs approximately 20 kg (45 lb). Move the FlowTrac unit by lifting, not sliding, to avoid scratches. Take care not to scratch the exterior surfaces of the cabinet while unpacking and lifting the unit. Take extra care not to damage the LCD and the keypad during handling.

Position the FlowTrac unit on the work surface (next to the LoadTrac-II load frame if it is a part of your system) so that there is easy access to the front and back panels. After electrical and hydraulic connections have been made to the back panel, the unit can be positioned so that only the front panel is easily accessible.

Attach the power cord, but do not plug it into an AC outlet yet.

A.3. Installing Network/Geonet-U

Your Geocomp system included a USB terminal to establish communication between the PC and your unit. Follow the instructions and images below to ensure the correct network connection.

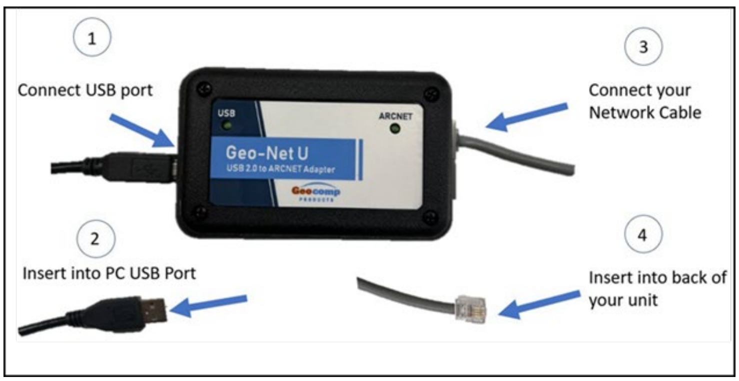


Figure A 1. USB 22 Adaptor Connections

- Connect the USB-C cable to the USB22 adaptor port (Figure A 1, Step 1).
- Insert the other end into one of your PC USB ports (Figure A 1, Step 2).
- Connect one end of your network cable to the USB22 adaptor (Figure A 1, Step 3).
- Connect the other end into the back of your Geocomp Unit (Figure A 2).
- Connect your terminator to the other port (Figure A 2). The terminator looks like a cut cable, do not throw it away.

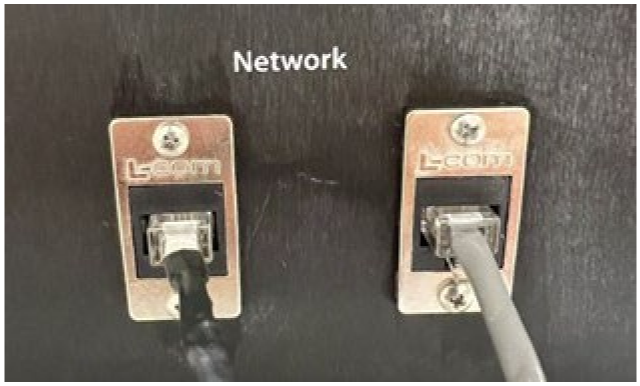


Figure A 2. Network Cable Connection

<u>*WARNING:</u> The network cable looks like a phone cable, but it is not. Do not use a regular phone cable. The system warranty is voided if anything other than a supplied cable is used.

A.4. Installing Control Software

Your Geocomp System includes .msi files for your Geocomp Software and the drivers for communication. They will be found on your USB flash drive (Software Folder). Run the *.msi files and restart your computer to install the software and drivers.

<u>*NOTE:</u> You will require administrative rights on the computer to install these files. A license is required to use the software, please check the software manual for details on how to request a license.

A.5. Boot Sequence

During the boot sequence, if the LCD screen seems to have a nonsensical display, if a flow pump motor starts to move and fails to stop or if there is any other indication of malfunction, turn off the main power switch on the FlowTrac back panel. This will completely power-down the FlowTrac unit and allow you to search for the cause of the problem. If Appendix C Troubleshooting does not help in finding the cause, call Geocomp Corp for assistance.

Turn on your computer and monitor (if they are not already on). Open a control program that you will use for running a test and load a test file.

Turn on the FlowTrac unit. The imbedded controller will go through a boot sequence which will take about fifteen seconds.

<u>*WARNING:</u> Do not press the Esc key on the keypad during the boot sequence. Your flow pump firmware has been configured at the factory for the conditions of your use. Pressing the Esc key during the boot sequence will remove this configuration.

At the start of the boot sequence, the Motor Power and CPU Power lights will glow a steady green while the Flow Out and Flow In lights will alternate rapidly on and off. The LCD panel will display the boot sequence message A (Figure A 3).

After a few seconds, the Flow Out and Flow In lights will turn off. A few seconds later, the LCD panel display will change to the boot sequence message B (Figure A 4).

Figure A 3. Boot Sequence Message A

Figure A 4. Boot Sequence Message B

Toward the end of the boot sequence, the Net Tx light will come on. All three LEDs in the left column must be a steady green for FlowTrac to function properly.

***<u>NOTE:</u> If the Net Tx LED is blinking, the network cable used to connect your computer to the load frame is probably not connected properly or a control program has not been opened. You may have to load a test file into your control program before the Net Tx LED will glow steadily. See your software manual for loading details.**

If the Net Tx LED is still blinking even though a control program is running (or one or more of the other LEDs is not glowing steadily), refer to Appendix C Troubleshooting or call Geocomp, Inc. for assistance.

At the completion of the boot sequence, the main menu (Figure A 5) will appear on the LCD screen.

Figure A 5. Main Menu

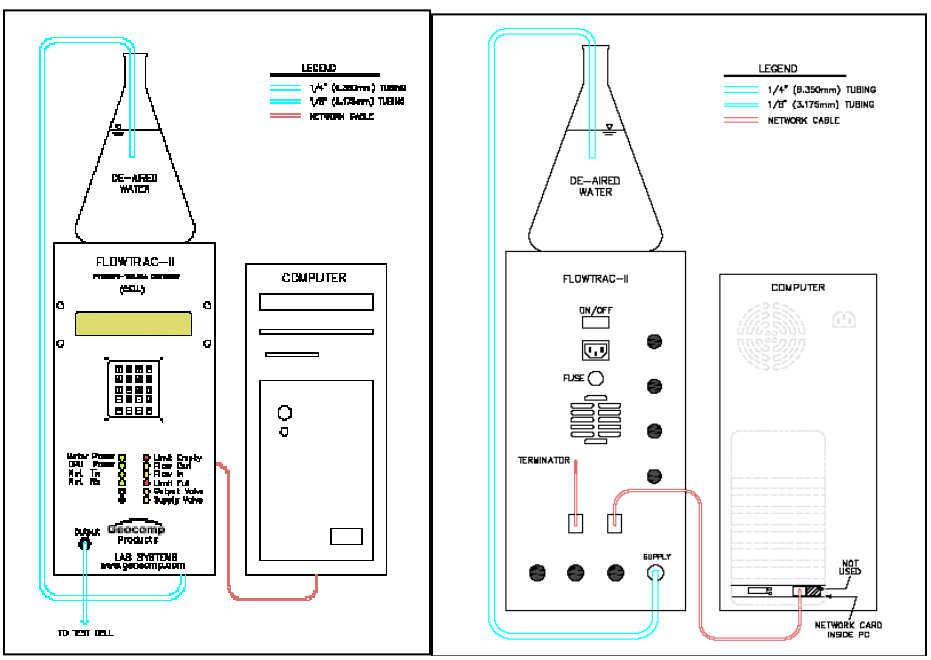
The ID number should be 66 if your application requires only one FlowTrac unit. (If your application requires two units, one should have the ID of 66 and the other 67 depending how the units are to be used. If your application requires three units, the IDs should be 66, 67 and 68.)

If you are using a power strip with a switch, you can use that switch to simultaneously turn on and off the entire FlowTrac system. But the Net Tx LED will not glow steadily until a control program is loaded.

A.6. FlowTrac System Hydraulic Connections

As shown in Figure A 6, there are two hydraulic ports. The Output port is located on the front of the unit and has a 1/8-inch fitting. It is used for connecting the flow pump to your test cell using 1/8" plastic tubing. If two FlowTrac units are being used, one is connected to the cell port and the other to the sample port of the test cell. Choose a length of 1/8-inch tubing (supplied with your FlowTrac system) that will easily reach from the FlowTrac unit to the location where you will place the test cell. Connect this tubing to the Output port on the front of the FlowTrac unit. Put a connector on the end of the tubing that will go to the test cell.

The Supply port is located on the back panel of the FlowTrac unit and has a ¼-inch push-in quick-connect fitting. It is used to either fill or empty the flow pump cylinder. Use ¼-inch plastic tubing (supplied with your FlowTrac system) to connect the Supply port to a source of de-aerated clean water that is positioned above the mid-height of the FlowTrac unit. The source can be a container, which sits on the top of the FlowTrac unit.



(A) Front View (B) Back View

Figure A 6. FlowTrac Hydraulic Connections Diagram

Before using the FlowTrac unit, the flow pump cylinder needs to be filled with clean, de-aerated water, and all air bubbles removed from the cylinder and tubing. FlowTrac Hydraulic Diagram for connections

***WARNING: Unless it is absolutely necessary, do not move the piston in the flow pump cylinder (by using the Fill, Empty, Initialize or Jog options on the LCD Position menu) unless there is water in the cylinder. Dry operation can damage the seal around the piston.**

***NOTE: If, while you are working through the following instructions, you suspect a malfunction or feel the need for an immediate emergency stop, turn off the power switch on the back of the FlowTrac unit. This will stop all movement of the flow pump and will not cause any damage to the pump or to the embedded controller. When the FlowTrac unit is turned on again, the controller will reboot and you can pick up where you left off.**

Place the free end of the tubing connected to the Supply port into an empty container. Select the Position option on the main menu by pressing 2 on the keypad to display the Position menu. Next, press 1 to select the Empty option and then press Ent. The Supply Valve LED will glow, showing that the supply valve is open, and the "Flow Out" LED will glow as the piston in the flow pump cylinder moves to the completely empty position. When this position is reached, the piston will stop and the "Limit Empty" LED will start blinking.

Move the tubing to your source of de-aerated water. Be sure that the end of the tubing is near the bottom of the container and is fastened so that it will not accidentally be pulled out of the water. Return to the Position menu on the LCD screen (press Esc twice), press 2 to select the Fill option, and then press Ent. The "Supply Valve" and "Flow In" LEDs will glow as the flow pump cylinder begins to fill with water. Let the fill process continue until the piston stops and the "Limit Full" LED starts blinking. Again return to the Position menu and use the keypad to empty the cylinder. You should see bubbles moving through the plastic tubing and coming out the end that is in the container of water. Cycle the fill-empty sequence several times until there are no more bubbles in the tubing.

At this point it is best to run water through the tubing that has been connected to the Output port and will connect the test cell. Be sure that the flow pump cylinder is not completely empty. Place the unconnected end of the tubing from the Output port into an empty container on the floor. Select the Control Valves option from the Position menu, close the Supply Valve and open the Output Valve. Select the Jog option from the Position menu. Press the 2 key and then the Up arrow key to start the motor running at 100 steps/sec. (see Section 2.1.2.4 Jog) Press the Up arrow key again and the motor will be running at 200 steps/sec. Soon you will see water moving through the tubing. When the water begins to run out into the container and there are no bubbles in the tubing, press the Esc key

Once you are sure that all of the air has been pumped out of the flow pump cylinder and out of both lengths of tubing, use the Initialize option to fill the flow pump cylinder to 50% capacity. Do this by selecting Initialize from the Position menu and then pressing Ent. The piston in the flow pump cylinder will empty the cylinder, activate the Empty limit switch, reverse direction, and then fill the cylinder to 50% capacity.

APPENDIX B: MAINTENANCE

There are no scheduled maintenance requirements for the FlowTrac equipment. However, we do have some recommendations (as described below) for keeping your equipment in the best possible working condition. The FlowTrac system comprises durable but precision components. As with any precision laboratory equipment, it should be operated with care and only by a skilled user. Have every person who will use the system become familiar with the functioning of the equipment through a study of this manual and the software manual, or through training by another skilled user. Strongly discourage the curious worker who drops by and wants to see by trial and error what the system will do.

B.1. FlowTrac Flow Pump

1. Clean the exterior of the FlowTrac unit with water and a mild dishwashing detergent. Do not use abrasives or any cleaner that reacts with aluminum, such as those containing ammonia. Do not use any abrasive scrubbing cloth.

2. Avoid water dripping into the interior of the FlowTrac unit. Electronics inside could be damaged.

3. Avoid sliding objects, which might cause scratches, over the FlowTrac case.

4. Do not remove the FlowTrac case unless you have been instructed to do so and know exactly what you are doing.

5. The fan slots on the back of the FlowTrac unit may collect a lot of dust. If this happens, turn off the FlowTrac power switch (on the back panel) so that the fan is off. Use a vacuum cleaner to carefully clean the dust from the slots.

6. The FlowTrac pump cylinder should be flushed after extensive use. Flushing is done by first emptying the cylinder. Use the keypad and LCD menus to do this (see Section 2.1.2.1 Empty). The cylinder must be emptied into a container separate from the one you fill from. After draining the cylinder, fill it with clean, de-aerated water. You should repeat this operation several times or until you can see that the drained water is clear.

7. The FlowTrac pump cylinder should be left empty if the unit will remain idle for a prolonged period of time.

B.2. FlowTrac Solenoid Valves

There are two solenoid valves; a supply valve on the back connected to a 6.35 mm (1/4 in) diameter tube, and an output valve on the front connected to a 3.17 mm (1/8 in) diameter tube. The supply valve is normally closed and the output valve is normally open. There are two conditions throughout all phases of all tests:

1. **Undrained condition:** The high-precision stepper motor keeps the volume constant by either increasing or decreasing the pressure to keep the volume at the same value. The increase/decrease in pressure is the sample pressure change during the undrained shear phase. The changes are related to the soil behavior of being contractive (positive excess pore pressure) or dilative (negative excess pore pressure). See Figure B 1.

2. **Drained condition:** Similarly the high-precision stepper motor keeps the pressure constant by either increasing or decreasing the volume to keep the pressure at the same value. The increase/decrease in volume is the sample volumetric change during the drained shear phase. The changes are related to the soil behavior of being contractive (positive volumetric strain) or dilative (negative volumetric strain). See Figure B 1.

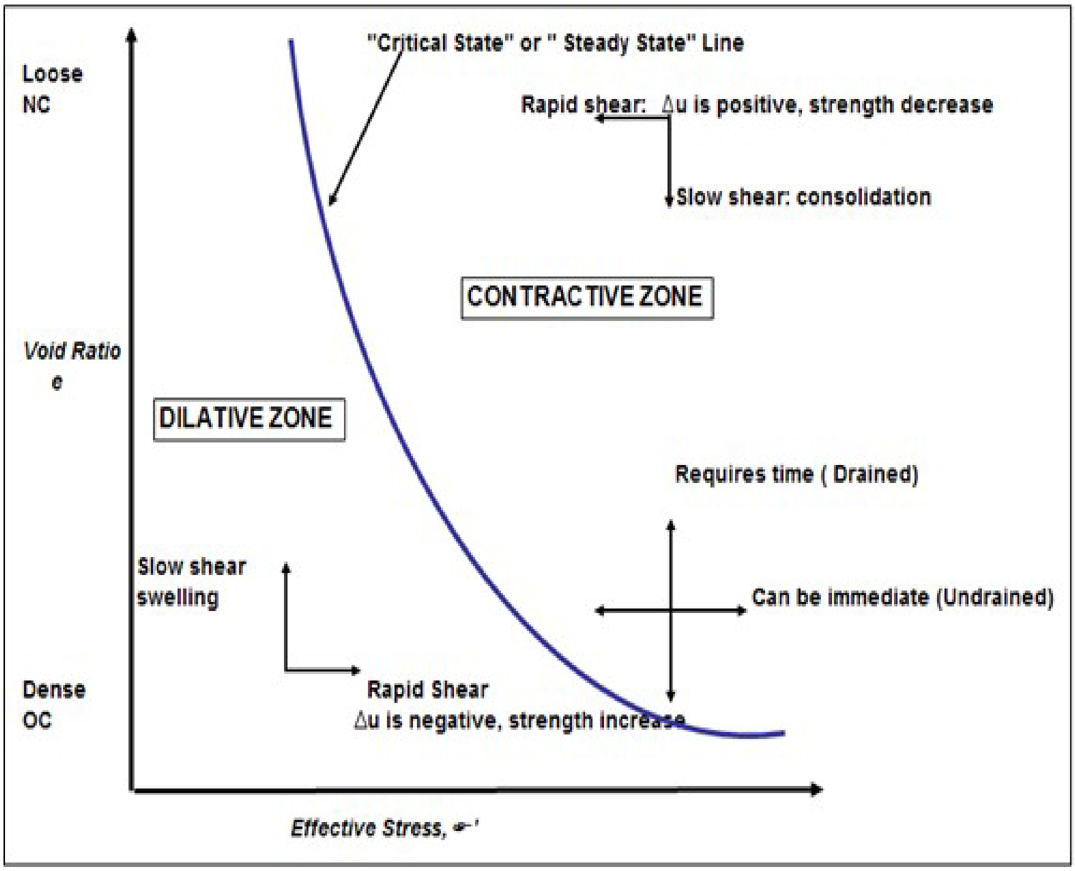


Figure B 1. Critical State Diagram

During a test, the stepper motor is always stepping one way or the other. Therefore the manual valves on the triaxial cell, the permeability cell or the Rowe cell must be always open; everything is happening at the location of the FlowTrac unit.

Occasionally the solenoid valves get clogged (see Figure B 2). If this happens, they need to be taken apart and flushed with a high-pressure air source. If this does not clear them, they should be replaced. To keep the valves from clogging, you should flush the pumps after each use by emptying and filling the pumps several times while watching the quality of the water coming out. Ideally, you should change the water after each flush to ensure that you are not getting the fine particles back into the pump.

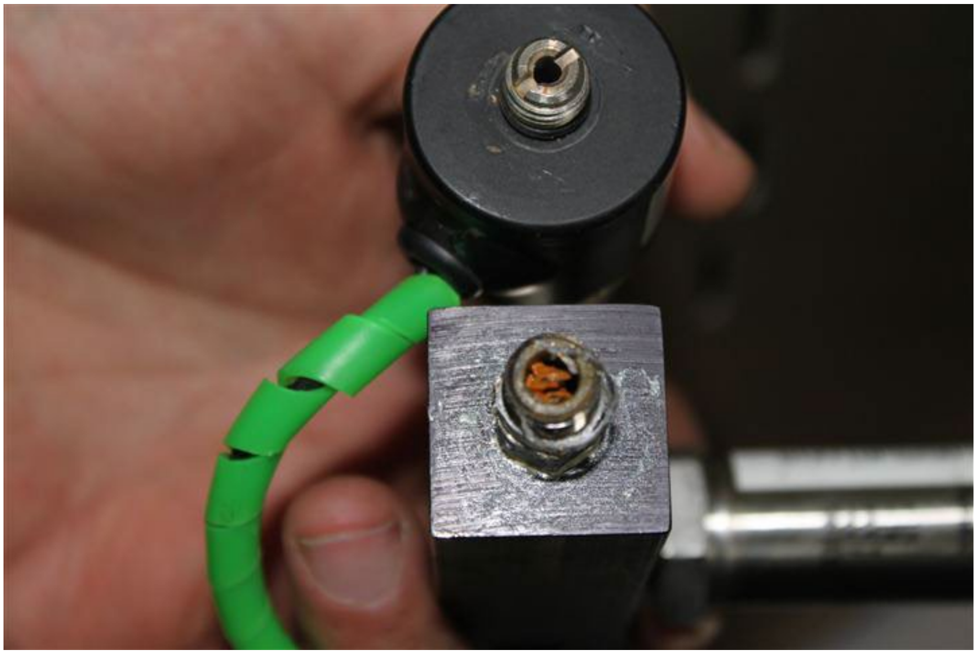


Figure B 2. Clogged Solenoid Valve

Supply Valve

To see if the supply valve is clogged, follow these steps:

1. Close the supply valve (using either keypad or software control).

2. Open the output valve.

3. Make sure that the 1/8" (3 mm) tube at the end of the output valve is connected to a valve that is closed so that you can apply and maintain a pressure within the tube.

4. Using software control, apply and maintain about 350 kPa (50 psi).

5. Check the flow pump activity. If the pump is not stabilizing because the pressure is dropping, then check to see if water is dripping out of the supply line tubing coming out of the back of the FlowTrac unit. (Clogging of the supply valve will result in it not sealing completely when closed.)

If water is dripping from the supply line, then for the time being you will need to use a manual valve on the back of the FlowTrac unit to close the line when a test is running. Open the valve only when you fill the pump with water at the beginning of a test.

Eventually you will have to replace the solenoid valve.

Output Valve

If the output valve is clogged, it will be obvious since no water will be coming out of the output line when the valve is open.

As a temporary measure you can remove the solenoid valves and replace them with manual valves. The manual output valve should always be open except when filling and emptying the pump; conversely the supply valve should always be closed except when filling and emptying the pump.

To summarize:

Filling the pump: Supply valve open and output valve closed.
Emptying the pump: Supply valve open and output valve closed.
Running a test: Supply valve closed and output valve open.

B.3. Computer

1. Keep the computer and monitor screen free of dust.

2. Clean the exterior of the computer with water and mild dishwashing detergent.

3. Avoid getting liquids inside the computer.

4. Avoid spilling liquids onto the keyboard.

5. In dusty environments, cover the keyboard when not in use.

APPENDIX C: TROUBLESHOOTING

If you experience difficulty while installing the FlowTrac system, the first step is to review the installation instructions carefully to see that you have followed each step correctly. If there is still a problem, look through the list below to see if any of the headings match your problem. If so try the suggested solution. If you continue to have trouble, contact Geocomp, Inc. for assistance.

"Memory checksum failed" messages when system is first booted:

Possible cause:

The program that was factory loaded into the non-volatile memory of the controller needs to be restored. Select the Setup option from the LCD main menu, then select each of the menus in the left column of Table C 1 and enter the parameters as shown in the table.

After you have entered all of the parameters, you will need to turn off the FlowTrac unit and turn it on again (or select the Restart option on the System Setup menu) for the change to become effective.

> ***WARNING:** After you have restored the parameters and your system is functioning properly, it is necessary to calibrate the pressure transducer. When you have done this, replace the Factor and Offset values shown above with the new values. At best, test will be inaccurate and at worst damage may occur if you do not do this.**

<table>
<tr><th>MENU</th><th>ITEM</th><th colspan="20">PARAMETERS</th></tr>
<tr><td rowspan="2">1. System</td><td>1. A/D Channel</td><td colspan="20">1 or 2 depending on intended FlowTrac use</td></tr>
<tr><td>2. Units</td><td colspan="20">Metric (or English if you prefer)</td></tr>
<tr><td>2. Network</td><td>1. Node ID</td><td colspan="20">66 (cell pressure unit), 67 (sample or bottom pressure unit), 68 (top pressure unit) (Can be changed to any value between 20 and 256.)</td></tr>
<tr><td rowspan="16">3. A/D</td><td rowspan="8">1. Channel 1</td><td colspan="5">1. Type</td><td colspan="5">2. Excitation</td><td colspan="5">3. Polarity</td><td colspan="5">4. Range</td></tr>
<tr><td colspan="5">Pressure</td><td colspan="5">12 V</td><td colspan="5">Bipolar</td><td colspan="5">160 mV</td></tr>
<tr><td colspan="5">5. Factor*</td><td colspan="5">6. Offset</td><td colspan="5">7. Low. Lim.</td><td colspan="5">8. Up. Lim.</td></tr>
<tr><td colspan="5">0.055 kPa/cnt</td><td colspan="5">32768 cnt</td><td colspan="5">0 cnt</td><td colspan="5">65535 cnt</td></tr>
<tr><td colspan="20">9. Advanced</td></tr>
<tr><td colspan="4">1. Mode</td><td colspan="4">2. Rate</td><td colspan="4">3. Filter</td><td colspan="4">4. Chop</td><td colspan="4">5. Fast</td></tr>
<tr><td colspan="4">Diff.</td><td colspan="4">200 Hz</td><td colspan="4">Enabled</td><td colspan="4">Enabled</td><td colspan="4">Enabled</td></tr>
<tr><td colspan="20"></td></tr>
<tr><td rowspan="8">2. Channel 2**</td><td colspan="5">1. Type</td><td colspan="5">2. Excitation</td><td colspan="5">3. Polarity</td><td colspan="5">4. Range</td></tr>
<tr><td colspan="5">Pressure</td><td colspan="5">12 V</td><td colspan="5">Bipolar</td><td colspan="5">160 mV</td></tr>
<tr><td colspan="5">5. Factor*</td><td colspan="5">6. Offset</td><td colspan="5">7. Low. Lim.</td><td colspan="5">8. Up. Lim.</td></tr>
<tr><td colspan="5">0.055 kPa/cnt</td><td colspan="5">32768 cnt</td><td colspan="5">0 cnt</td><td colspan="5">65535 cnt</td></tr>
<tr><td colspan="20">9. Advanced</td></tr>
<tr><td colspan="4">1. Mode</td><td colspan="4">2. Rate</td><td colspan="4">3. Filter</td><td colspan="4">4. Chop</td><td colspan="4">5. Fast</td></tr>
<tr><td colspan="4">Diff.</td><td colspan="4">200 Hz</td><td colspan="4">Enabled</td><td colspan="4">Enabled</td><td colspan="4">Enabled</td></tr>
<tr><td colspan="20"></td></tr>
<tr><td rowspan="8">4. PID</td><td colspan="6">1. P-Gain</td><td colspan="5">2. I-Gain</td><td colspan="5">3. D-Gain</td><td colspan="5">4. V-Offset</td></tr>
<tr><td colspan="6">3 steps/sec/cnt</td><td colspan="5">2 steps/sec/cnt</td><td colspan="5">0 steps/sec/cnt</td><td colspan="5">0 steps/sec</td></tr>
<tr><td colspan="6">5. I-Limit</td><td colspan="5">6. V-Limit</td><td colspan="5">7. Dither Ampl.</td><td colspan="5">8. Dither Freq.</td></tr>
<tr><td colspan="6">1024 steps/sec</td><td colspan="5">32768 steps/sec</td><td colspan="5">0 steps/sec</td><td colspan="5">0 Hz</td></tr>
<tr><td colspan="21">9. Advanced</td></tr>
<tr><td colspan="6">1. Filter</td><td colspan="5">2. FB Channel</td><td colspan="5">3. Anti-Windup</td><td colspan="5">4. Derivative FB</td></tr>
<tr><td colspan="6">256 Hz</td><td colspan="5">Channel 0</td><td colspan="5">Reset</td><td colspan="5">De/dt</td></tr>
<tr><td colspan="21"></td></tr>
<tr><td rowspan="4">5. Motor</td><td>1. Frequency</td><td colspan="20">16384 Hz</td></tr>
<tr><td>2. Acceleration</td><td colspan="20">40960 steps/sec/sec</td></tr>
<tr><td>3. Step Capacity</td><td colspan="20">456654</td></tr>
<tr><td>4. Step factor</td><td colspan="20">5.4x10^{-4} cc/step, 250 cc unit; 1.7x10^{-3} cc/step, 750 cc unit</td></tr>
</table>

*The factor shown is approximate. It is unique to each sensor and needs to be updated after each calibration procedure.
**The settings shown for Channel 2 are those for when a FlowTrac unit is used for Triaxial tests. If the unit will be used for the cell pressure in Rowe Consolidation tests, Channel 2 will be set up for a displacement transducer. See Table C 2 below for setup parameters.

Table C 1. Summary of FlowTrac Settings

Channel 2 Configuration for Rowe Consolidation Tests						
FlowTrac Use	Channel 2 use	Settings				
Cell pressure	Vertical Displacement	1. Type	2. Excitation	3. Polarity	4. Range	
		Displacement	5 V	Unipolar	2560	
		5. Factor*	6. Offset	7. Low. Lim.	8. Up. Lim.	
		8.0×10^{-4} mm/cnt	32768 cnt	0 cnt	65535 cnt	
		9. Advanced				
		1. Mode	2. Rate	3. Filter	4. Chop	5. Fast
		Diff.	200 Hz	Enabled	Enabled	Enabled
Sample pressure	External Sample Pressure	1. Type	2. Excitation	3. Polarity	4. Range	
		Pressure	12 V	Bipolar	160 mV	
		5. Factor*	6. Offset	7. Low. Lim.	8. Up. Lim.	
		0.055 kPa/cnt	32768 cnt	0 cnt	65535 cnt	
		9. Advanced				
		1. Mode	2. Rate	3. Filter	4. Chop	5. Fast
		Diff.	200 Hz	Enabled	Enabled	Enabled

*The factor shown is approximate. It is unique to each sensor and needs to be updated after each calibration procedure.

Table C 2. Summary of Rowe Consolidation Test Settings

Net Tx LED is blinking:
and/or
FlowTrac cannot be controlled using the computer control software:
and/or
"Error initializing network adapter" when starting test with software:

Possible causes:

1. Probably the USB Arcnet Service window was closed or had an error. If it tells you to Unplug your Geonet-U, just remove the black cable, both lights in the Geonet-U adaptor should turn off. Re connect the cable and open your software, the window should look like the picture below. **Do not close this window, as you will lose network communication.**

2. The network communication cable has not been connected to the computer or it is plugged into the Ethernet jack on the back of your PC instead of into the special (black) adapter that is plugged into the network communication card provided by Geocomp, Inc.

3. The Node ID on the LoadTrac-III LCD Node ID entry screen does not match the Node ID on the Hardware Setup window of the controlling software. See Section 2.1.2.1 Empty for the value that should be in the LoadTrac-III imbedded controller and then check the value in the Hardware Setup window.

Computer (test control software) not communicating with the FlowTrac unit:

Possible causes:

1. The Node ID setting on the FlowTrac unit is not the same as the value on the control software. Check to be sure the Node ID value (option 1 on the System Setup menu) is the same as the appropriate ID on the Hardware Setup menu shown when Hardware is selected from the Options menu.

2. Network cable is loose or unplugged. Check to make sure that the network cable is firmly connected to both the back panel of the FlowTrac unit and the computer, and that the terminator cable is connected to the back panel of the load frame.

LCD screen is blank:

Possible causes:

1. The cable connecting the LCD to the Controller board inside the FlowTrac unit may not be connected properly, or may have come loose during shipping.

2. The LCD screen may have been damaged during shipping.

3. The LCD screen may have become defective.

Motor stalls and makes grinding noise:

Possible causes:

1. The flow pump is trying to produce an excessive pressure and the motor is not capable of further increasing the pressure. Calibrate the channel 1 pressure sensor (your software provides an easy calibration procedure) and compare the results to the calibration factor stored in the embedded controller (option 5 on the A/D Channel 1 Setup menu described in Section 2.1.4.3 A/D (Analog to Digital)). If there is a difference, enter the new correct value. (Also, be sure that the calibration factor on the software Calibration Summary window is correct.)

2. The frequency setting may be too high. Check the Frequency (option 1 on the Motor Setup menu described in Section 2.1.4.2 Network). The value should be 16384 Hz. If the setting is too high, enter the correct value.

Flow pump motor fails to run:

Possible causes:

1. There is no power to either the motor or the embedded controller. Check to be sure that the two power lights (Motor and CPU) on the front of the FlowTrac unit are lit.

2. The P-Gain value (option 1 on the LCD PID Setup menu) is zero. The value should be 1. If the value needs to be reset, enter the correct value. (Also, be sure that the P-Gain value on the software Load PID Settings window is not zero.)

3. The Frequency value (option 1 on the Motor Setup menu described in Section 2.1.4.5 Motor) is too high. The value should be no higher than 16384 Hz. If it needs to be reset, enter the correct value.

4. The V-Limit value (option 6 on the LCD PID Setup menu) is zero. The value should be 32768. If it needs to be reset, enter the correct value. (Also, be sure that the V-limit value on the software Load PID Settings window is not zero.)

Only one channel active for cyclic triaxial or Rowe consolidation testing:

Possible cause:

The AD Channels setting is 1 instead of 2 (option 2 on the System Setup menu).

The firmware default setting for a FlowTrac unit is one channel. When the FlowTrac unit is used for cyclic triaxial or Rowe consolidation testing the unit has to be configured to read two channels. If the Esc key is pressed during the boot sequence by mistake, the FlowTrac unit reverts to the default setting. If this has happened to your unit, go through the following steps to change the number of channels from 1 to 2 and then configure channel 2 for the appropriate use.

1. Turn on the FlowTrac unit. There will be two screen displays (Figure C 1) during the boot process. Watch the LCD and make sure that the system boots with no error messages. Do not press the Esc key on the keypad while the control board is booting.

```
FlowTrac II.          Rev: xx.xx.xxxx          ID: #
Copyright (c) GEOCOMP Corp. 1997-20xx
All rights reserved.
Performing system test. Please wait...
```

```
FlowTrac II.          Rev: xx.xx.xxxx          ID: #
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```

Figure C 1. Boot Sequence Messages

After about 15 seconds, the main menu will appear.

2. With the main menu displayed (Figure C 2), press keypad 4 to select Setup.

```
FlowTrac II.          Rev: xx.xx.xxxx          ID: #
1. Monitor            4. Setup
2. Position
3. Control
```

Figure C 2. Main Menu

3. With the Setup menu displayed (Figure C 3), press keypad 1 to select System.

```
Setup
1. System             4. PID
2. Motor
3. A/D
```

Figure C 3. Setup Menu

4. With the System Setup menu displayed (Figure C 4), press keypad 2 to select A/D Channels.

```
System Setup
1. Node ID
2. A/D Channels
3. Units
```

Figure C 4. System Setup Menu

5. The A/D Channel Count entry screen will show the number 1 if only one, instead of two channels is in use. Press keypad 2. Make sure the 1 changes to 2. Press the Ent key. Press the Esc key.

6. Shut off the FlowTrac unit and then turn it back on. Do not press Esc during the boot process.

7. Again display the Setup menu. This time select AD (option 3).

8. The AD Setup menu (Figure C 5) should show two channels.

Figure C 5. A/D Setup Menu

Press keypad 2 to select Channel 2.

9. On the AD Channel 2 Setup menu (Figure C 6), select each item and enter the appropriate parameter as shown in the tables of parameters at the beginning of Appendix C Troubleshooting.

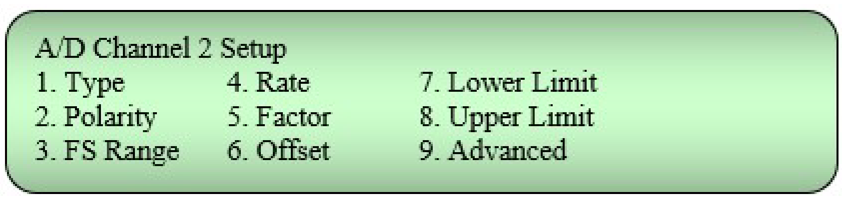

Figure C 6. A/D Channel 2 Setup Menu

10. After you have finished entering all the parameters, Shut off the FlowTrac unit and then turn it back on. Do not press Esc during the boot process.

11. Quick check: With the main menu showing on the LCD, select Monitor and then select System. You should see readings for 2 channels displayed on the LCD. If there are no sensors connected, the channel readings will float.

IMPORTANT NOTES:

Rowe consolidation setup:

On the FlowTrac unit that controls the cell pressure, the second channel is configured for the vertical displacement sensor. On the FlowTrac unit that controls the sample pressure, the second channel is configured for the external sample pressure sensor.

Cyclic triaxial setup:

On the FlowTrac unit that controls the cell pressure, the second channel is configured for the external cell pressure sensor. On the FlowTrac unit that controls the sample pressure, the second channel is configured for the external sample pressure sensor.

APPENDIX D: FREQUENTLY ASKED QUESTIONS

Q: What is the sustainable maximum pressure the flow pumps can maintain?

A: 200 psi.

Q: How much water should be used to fill the pumps prior to a test? Since there is a need to accommodate inflow from the sample during the consolidation, I assume the pump should have sufficient capacity (volume) to accommodate inflow of water. Is the pump being filled (prior the test) only halfway?

A: The maximum capacity of the FlowTrac unit is 250 cc. The use of the software option to initialize the flow pumps will fill the pumps to the volume specified on the Hardware Setup window (Hardware item on the Options menu). Typical settings are 75% for the cell pump and 50% for the sample pump. If your sample is fairly porous, you will need to change these settings to as high as 90% for the sample pump and as low as 10% for the cell pump. If you need to change the settings during a test, click Cancel, make the changes on the Hardware Setup window and restart the test.

Q: Is there any chance that during the test the pump will take in water from the flask, so the water has to be exposed to the atmospheric pressure (open to air).

A: No. During the test, your supply valve will be closed. But you may need to refill your sample flow pump during the test if your sample takes more than the volume of water initially in your pump.